Limits and Continuity

Limits

Let's begin with an *informal definition of limit*, postponing the precise definition until we've gained more insight.

Let $f(x)$ be defined on an open interval about x_0, *except possibly* at x_0 itself. If $f(x)$ gets arbitrarily close to L (as close to L as we like) for all x sufficiently close to x_0, we say that f approaches the **limit** L as x approaches x_0 and we write

$$\lim_{x \to x_0} f(x) = L,$$

Which is read as "the limit of $f(x)$ as x approaches x_0 is L".

This definition is "informal" because phrases like *arbitrarily close* and *sufficiently close* are imprecise; their meaning depends on the context. To a machinist manufacturing a piston, *close may mean within a few thousandths of an inch*. To an astronomer studying distant galaxies, *close may mean within a few thousand light-years*. We will need the ***precise definition***, however, when we set out to prove theorems about limits.

Example. *Behavior of a Function Near a Point*

How does the function $f(x) = \frac{x^2-1}{x-1}$ behave near $x = 1$?

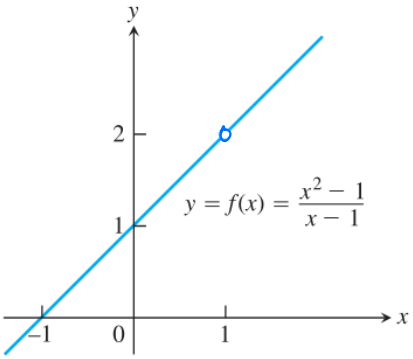
Solution:

$$f(x) = \frac{x^2-1}{x-1}, \quad x \neq 1$$

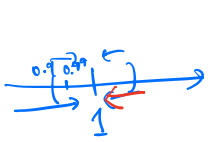
$$= \frac{(x-1)(x+1)}{x-1} \qquad x \neq 1$$

$$= x+1$$

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} x+1 = 2.$$



Note that $f(1)$ is not defined, it is clear that we can make the value of $f(x)$ *as close as we want* to 2 by choosing x *close enough* to 1 (in the following table).



$$\lim_{x\to 1}\frac{x^2-1}{x-1} = ?$$

The closer x gets to 1, the closer $f(x) = (x^2 - 1)/(x - 1)$ seems to get to 2

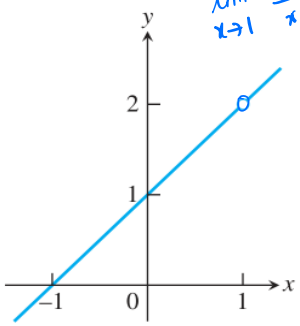
Values of x below and above 1	$f(x) = \dfrac{x^2-1}{x-1} = x + 1, \quad x \neq 1$
0.9	1.9
1.1	2.1
0.99	1.99
1.01	2.01
0.999	1.999
1.001	2.001
0.999999	1.999999
1.000001	2.000001

$$\lim_{x\to 1}\frac{x^2-1}{x-1} = 2.$$

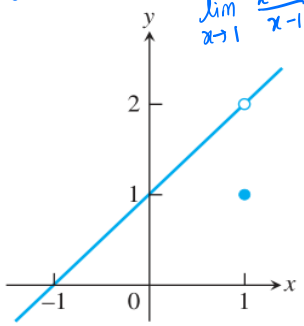
We say that $f(x)$ approaches the *limit* 2 as x approaches 1, and write as

$$\lim_{x\to 1} f(x) = 2, \text{or } \lim_{x\to 1}\frac{x^2-1}{x-1} = 2.$$

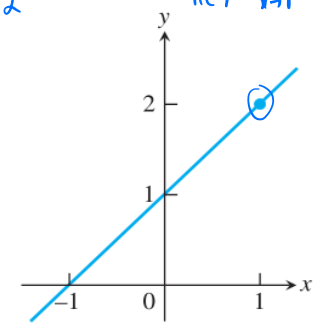
Example. *The Limit Value Does Not Depend on How the Function Is Defined at x_0.*



(a) $f(x) = \dfrac{x^2-1}{x-1}$, $x \neq 1$



(b) $g(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$



(c) $h(x) = x + 1$ $\quad h(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$

The function f above has limit 2 as $x \to 1$ even though f is not defined at $x = 1$. The function g has limit 2 as $x \to 1$ even though $g(1) \neq 2$. The function h is the only one whose limit as $x \to 1$ equals its value at $x = 1$. Note, $\lim_{x\to 1} h(x) = h(1)$.

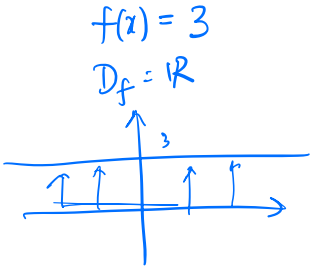
Sometimes $\lim_{x\to x_0} f(x)$ can be evaluated by calculating $f(x_0)$. This holds, for example, whenever $f(x)$ is an algebraic combination of polynomials and trigonometric functions for which $f(x_0)$ is defined.

Example.

(a) $\lim_{x\to 1} 4 = 4$ (b) $\lim_{x\to 2}(5x - 3) = 5(2) - 3 = 7$

(c) $\lim_{x\to 3} x = 3$ (d) $\lim_{x\to -2} \frac{3x+4}{x+5} = \frac{-6+4}{-2+5} = -\frac{2}{3}$

Example. *The Identity and Constant Functions Have Limits at Every Point*



(a) If f is the **identity function** $f(x) = x$, then for any value of x_0,

$$\lim_{x\to x_0} f(x) = \lim_{x\to x_0} x = x_0.$$

(b) If f is the **constant function** $f(x) = k$, (function with the constant value k), then for any value of x_0,

$$\lim_{x\to x_0} f(x) = \lim_{x\to x_0} k = k.$$

Some ways that limits can fail to exist, see the figure in the following.

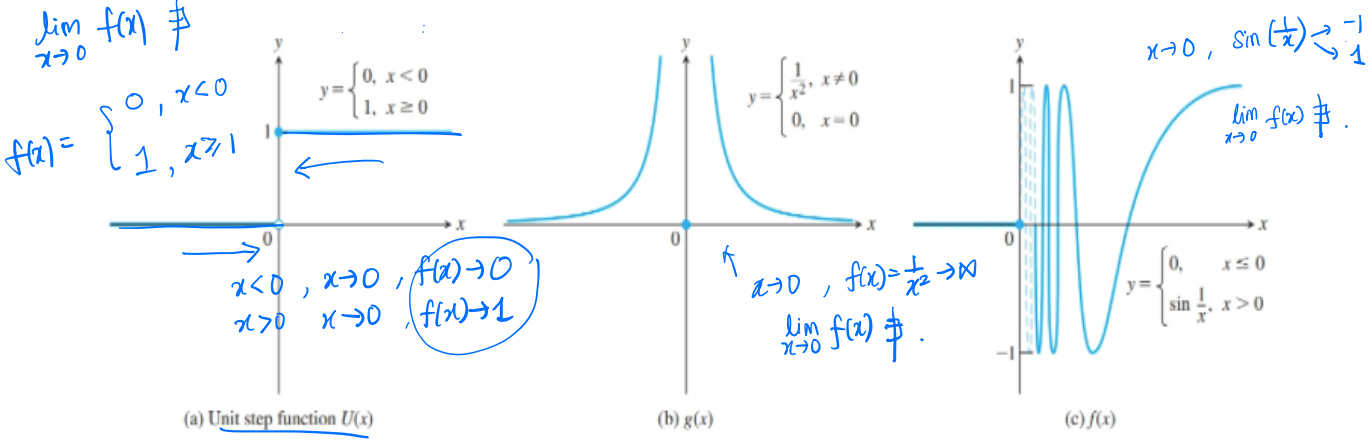


FIGURE 2.10 None of these functions has a limit as x approaches 0 (Example 4).

Example. *A Function May Fail to Have a Limit at a Point in Its Domain*

Discuss the behavior of the following functions as $x \to 0$.

(a) $U(x) = \begin{cases} 0, & x < 0 \\ 1. & x \geq 0 \end{cases}$

It *jumps*: The **unit step function** $U(x)$ has no limit as $x \to 0$ because its values jump at $x = 0$. For negative values of x arbitrarily close to zero, $U(x) = 0$. For positive values of x arbitrarily close to zero, $U(x) = 1$. There is no *single* value L approached by $U(x)$ as $x \to 0$.

(b) $g(x) = \begin{cases} \frac{1}{x^2}, & x \neq 0 \\ 0. & x = 0 \end{cases}$

It *grows too large to have a limit*: $g(x)$ has no limit as $x \to 0$ because the values of g grow arbitrarily large in absolute value as $x \to 0$ and do not stay close to *any* real number. We say g is not bounded.

(c) $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin\frac{1}{x}. & x > 0 \end{cases}$

It *oscillates too much to have a limit*: $f(x)$ has no limit as $x \to 0$ because the function's values oscillate between +1 and -1 in every open interval containing 0. The values do not stay close to *any* one number as $x \to 0$.

The problem with the function f discussed is not that it oscillates, but that it oscillates too much for a limit to exist.

A function that oscillates may or may not have a limit. In *Example later* we will see a function that oscillates wildly, but nevertheless does have a limit.

The Limit Laws

A few basic rules allow us to break down complicated functions into simple ones when calculating limits. By using these laws, we can greatly simplify many limit computations.

THEOREM 1—Limit Laws

If L, M, c, and k are real numbers and

$$\lim_{x \to c} f(x) = L \quad \text{and} \quad \lim_{x \to c} g(x) = M, \quad \text{then}$$

1. *Sum Rule:* $\lim_{x \to c} (f(x) + g(x)) = L + M$
2. *Difference Rule:* $\lim_{x \to c} (f(x) - g(x)) = L - M$
3. *Constant Multiple Rule:* $\lim_{x \to c} (k \cdot f(x)) = k \cdot L$
4. *Product Rule:* $\lim_{x \to c} (f(x) \cdot g(x)) = L \cdot M$
5. *Quotient Rule:* $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$
6. *Power Rule:* $\lim_{x \to c} [f(x)]^n = L^n$, n a positive integer
7. *Root Rule:* $\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}$, n a positive integer

(If n is even, we assume that $f(x) \geq 0$ for x in an interval containing c.)

Example. Evaluate the limit

(a) $\lim_{x \to c} (x^3 + 4x^2 - 3)$

(b) $\lim_{x \to c} \dfrac{x^4 + x^2 - 1}{x^2 + 5}$

(c) $\lim_{x \to -2} \sqrt{4x^2 + 3}$

Evaluating Limits of Polynomials and Rational Functions

Theorem. *Limits of Polynomials Can Be Found by Substitution*

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, then

$$\lim_{x \to c} P(x) = P(c) = a_n c^n + a_{n-1} c^{n-1} + \cdots + a_0.$$

Theorem. *Limits of Rational Functions Can Be Found by Substitution If the Limit of the Denominator Is Not Zero.*

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \to c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}$$

$Q(c) \neq 0$.

Note: This theorem applies only if the denominator of the rational function is not zero at the limit point c.

Example. Limit of a Rational Function $\lim_{x \to -1} \frac{x^3+4x^2-3}{x^2+5}$

$(-1)^2+5 = 6 \neq 0$

$\lim_{x \to -1} \frac{x^3+4x^2-3}{x^2+5} = \frac{(-1)^3+4(-1)^2-3}{(-1)^2+5} = \frac{0}{6} = 0.$

Example. Evaluate $\lim_{x \to 2} \frac{x^2+4x-12}{x^2-3x}$.

$4-6=-2 \neq 0$

$\lim_{x \to 2} \frac{x^2+4x-12}{x^2-3x} = \frac{2^2+4(2)-12}{-2} = \frac{0}{-2} = 0.$

If the denominator is zero, canceling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at c. If this happens, we can find the limit by substitution in the simplified fraction.

$P(x) = 2x^3+2x^2+x+1 = (x+1)(2x^2+1)$

$P(-1) \neq 0$ $(x+1)$

$2x^3+2x^2+x+1$

$= (x+1)(2x^2+0x+1)$

$= (x+1)(2x^2+1)$

$2x^2+1$

$x+1 \overline{)\, 2x^3+2x^2+x+1}$

$(-)\ 2x^3+2x^2$

$x+1$

Eliminating Common Factors from Zero Denominators

Example. *Cancelling the Common Factor*

Find $\lim_{x \to 1} \frac{x^2-1}{x-1}$.

$\lim_{x \to 1} \frac{x^2-1}{x-1} \to \frac{0}{0}$

$= \lim_{x \to 1} \frac{(x-1)(x+1)}{x-1}$

$= \lim_{x \to 1} x+1 = 2.$

Solution:

We can't find the limit by plugging in $x = 1$ because it makes the denominator zero. However, notice that

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1$$

after cancelling the common factor $x-1$. We able to cancel the common factor because when we take the limit as x approaches 1, $x \neq 1$, and so $x-1 \neq 0$.

$$\lim_{x \to 1} \frac{x^2-1}{x-1} = \lim_{x \to 1} (x+1) = 2.$$

Example. Evaluate $\lim_{h\to 0}\frac{2(h-3)^2-18}{h}$.

Solution

$$\lim_{h\to 0}\frac{2(h-3)^2-18}{h}=\lim_{h\to 0}\frac{2(h^2-6h+9)-18}{h}=\lim_{h\to 0}\frac{2h^2-12h+18-18}{h}$$

$$=\lim_{h\to 0}\frac{2h^2-12h}{h}=\lim_{h\to 0}(2h-12)=-12.$$

Example. *Creating and Cancelling a Common Factor*

Find $\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}$.

Solution

We can't plug in $t=0$, and the numerator and denominator have no obvious common factors. We can create a common factor by multiplying both numerator and denominator by the expression $\sqrt{t^2+9}+3$ (obtained by changing sign after the square root).

$$\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}=\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}\cdot\frac{\sqrt{t^2+9}+3}{\sqrt{t^2+9}+3}$$

$$=\lim_{t\to 0}\frac{t^2+9-9}{t^2\left(\sqrt{t^2+9}+3\right)}$$

$$=\lim_{t\to 0}\frac{1}{\sqrt{t^2+9}+3}$$

$$=\frac{1}{6}$$

Exercises:

1. $\lim_{x\to 3}\frac{x^2-2x-3}{x^2-5x+6}$

2. $\lim_{x\to -2}\frac{\sqrt{2-x}-2}{x+\sqrt{6+x}}$

3. $\lim_{x\to 0}\frac{x}{\sqrt{x+4}-2}$

The Sandwich Theorem

It is called the Sandwich Theorem because it refers to a function f whose values are sandwiched between the values of two other functions g and h that have the same limit L at a point c.

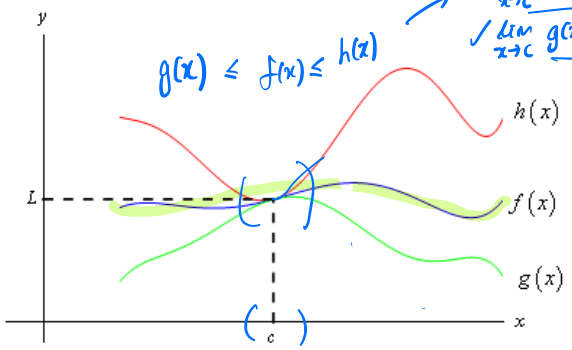
Theorem. The Sandwich Theorem (Squeeze Theorem or Pinching Theorem)

Suppose that $g(x) \le f(x) \le h(x)$ for all x in some open interval containing c, except possibly at $x = c$ itself. Suppose also that

$$\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L.$$

Then $\lim_{x\to c} f(x) = L$.

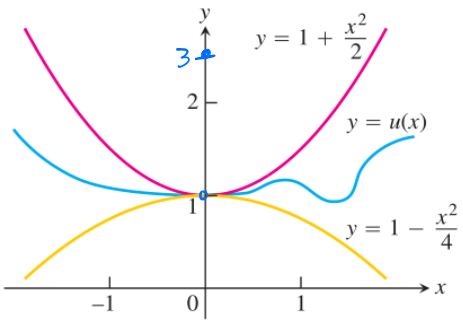
If we assume that all three functions f, g, and h are continuous (not required by the theorem), then following figure illustrates what is happening with the Sandwich Theorem.



Example. *Applying the Sandwich Theorem*

Given that $1 - \frac{x^2}{4} \le u(x) \le 1 + \frac{x^2}{2}$ for all $x \neq 0$, find $\lim_{x\to 0} u(x)$, no matter how complicated u is.

Solution.
Refer to the figure below.



Example. Evaluate $\lim_{x\to 0} x^2 \cos\frac{1}{x}$.

Solution.

$-1 \le \cos\theta \le 1$

$x=0$

$f(x) = x^2\cos\frac{1}{x}$ $\quad -1 \le \cos\frac{1}{x} \le 1$ $\quad x^2>0$, $x\neq 0$

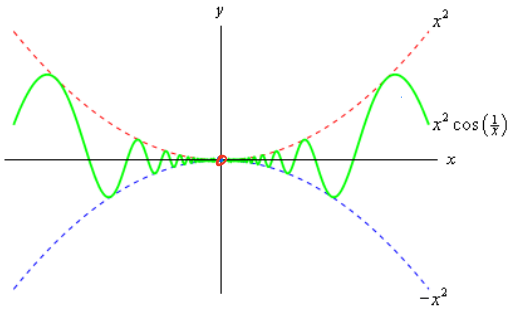
$-x^2 \le x^2\cos\frac{1}{x} \le x^2$

$\lim_{x\to 0}(-x^2) = 0$

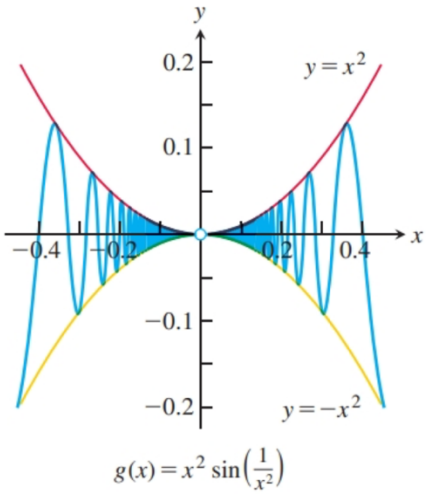
$\lim_{x\to 0}(x^2) = 0$

By Sandwich theorem,

$\lim_{x\to 0} x^2\cos\frac{1}{x} = 0$



Example. How does the function $g(x) = x^2 \sin\left(\frac{1}{x^2}\right)$ behave near $x = 0$?



$g(x) = x^2 \sin\left(\frac{1}{x^2}\right)$

$-1 \le \sin\frac{1}{x^2} \le 1$

$-x^2 \le x^2\sin\frac{1}{x^2} \le x^2$

$\lim_{x\to 0} -x^2 = 0$

$\lim_{x\to 0} x^2 = 0$

By Sandwich theorem

$\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right) = 0$.

$-|\theta| \le \sin\theta \le |\theta|$ $\quad \lim_{\theta\to 0}|\theta| = 0$ By Sandwich theorem

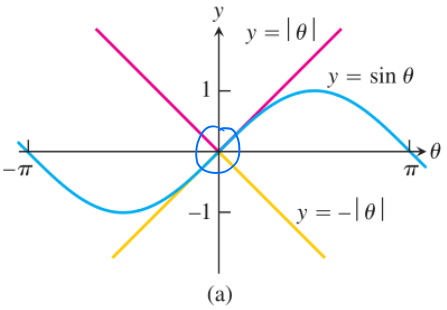
$\lim_{\theta\to 0}\sin\theta = 0$ $\quad \lim_{\theta\to 0}-|\theta| = 0$ $\quad \lim_{\theta\to 0}\sin\theta = 0$

Example. *More Applications of the Sandwich Theorem*

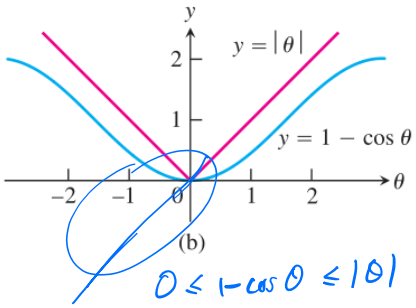
(a) Follows from the definition of $\sin\theta$ that $-|\theta| \le \sin\theta \le |\theta|$ for all θ, and since $\lim_{\theta\to 0}(-|\theta|) = \lim_{\theta\to 0}(|\theta|) = 0$, we have

$$\lim_{\theta\to 0}\sin\theta = 0.$$

$\sin\theta \simeq \theta$ θ small



(a)



(b)

$0 \le 1-\cos\theta \le |\theta|$

$\theta \simeq$ small $\quad \cos\theta \simeq 1-\frac{\theta^2}{2}$. $\quad \theta\to 0$ $\quad 0 < 1-\cos\theta \simeq \frac{\theta^2}{2} < |\theta|$ $\quad \theta\to 0 \to 0<\theta<1$ $\quad \theta^2<\theta$

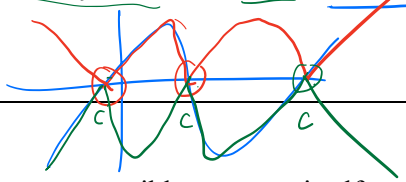
(b) Follows from the definition of $\cos\theta$ that $0 \le 1-\cos\theta \le |\theta|$ for all θ, hence,

$$\lim_{\theta\to 0}1-\cos\theta = 0 \quad \text{or} \quad \lim_{\theta\to 0}\cos\theta = 1$$

(c) For any function $f(x)$, if $\lim_{x\to c}|f(x)| = 0$, then $\lim_{x\to 0}f(x) = 0$.

(The argument: $-|f(x)| \le f(x) \le |f(x)|$ and $-|f(x)|$ and $|f(x)|$ have limit 0 as $x \to c$)

$\lim_{x\to c}g(x) = \lim_{x\to c}h(x) = L$ $\quad -|f(x)| \le f(x) \le |f(x)|$



Theorem.

If $f(x) \le g(x)$ for all x in some open interval containing c, except possibly at $x = c$ itself, and the limits of f, g both exist as x approaches c, then

$$\lim_{x\to c}f(x) \le \lim_{x\to c}g(x).$$

$f(x) \le g(x)$

$\lim_{x\to c}g(x) = L$

$\Rightarrow \lim_{x\to c}f(x) \le L$

The Precise Definition of a Limit

We replace phrases like "gets arbitrarily close to" in the informal definition with specific conditions that can be applied to any particular example.

To show that the limit of $f(x)$ as $x \to x_0$ equals the number L, we need to show that the gap between $f(x)$ and L can be made "as small as we choose" if x is kept "close enough" to x_0. Let us see what this would require if we specified the size of the gap between $f(x)$ and L.

Example. *A Linear Function*

Consider the function $y = 2x - 1$ near $x_0 = 4$. Intuitively, it is clear that y is close to 7 when x is close to 4, so $\lim_{x\to 4}(2x-1) = 7$. However, how close to $x_0 = 4$ does x have to be so that $y = 2x - 1$ differs from 7 by, say less than 2 units?

$y = 2x-1$, $x_0 = 4$, $y = 7$.

$y, 7 \leftrightarrow$ 2units.

$\forall \varepsilon > 0$, exist $\delta > 0$

$0 < |x-a| < \delta \Rightarrow |f(x)-L| < \varepsilon$

input $\quad$ output

$|x-4| < \delta \Rightarrow |y-7| < 2$ $\quad y = 2x-1$

if p, then q.

$\delta = 1$

$|y-7|<2 \Rightarrow |2x-1-7|<2$

$\Rightarrow |2x-8|<2$

$\Rightarrow 2|x-4|<2$

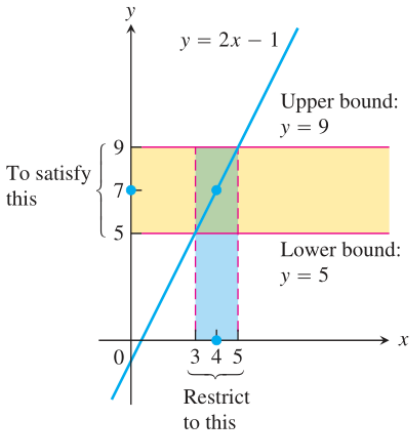
$|x-4|<1$

Let $|y-7|<2$, then

$|x-4|<1 \Rightarrow |2x-1-7| = |2(x-4)|$

$= 2|x-4| < 2(1) \Rightarrow$

Solution.



We are asked: For what values of x is $|y-7|<2$?

To find the answer, we first express $|y-7|$ in terms of x:

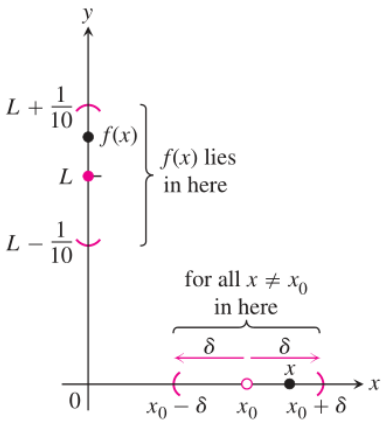
$$|y-7| = |(2x-1)-7| = |2x-8|.$$

The question then becomes: what values of x satisfy the inequality $|2x-8|<2$?

To find out, solve the inequality:

$$\begin{aligned} &|2x-8|<2 \\ &-2<2x-8<2 \\ &6<2x<10 \\ &3<x<5 \\ &-1<x-4<1. \end{aligned}$$

Keeping x within 1 unit of $x_0 = 4$ will keep y within 2 units of $y_0 = 7$.

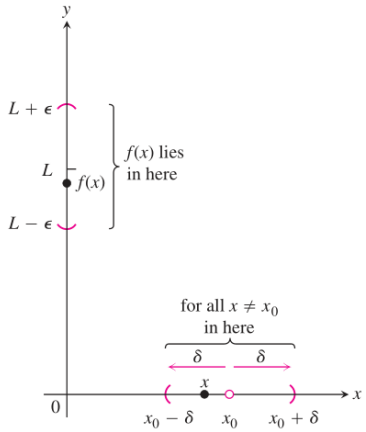


Suppose we are watching the values of a function $f(x)$ as x approaching x_0 (without taking x_0 itself).

Certainly, we want $f(x)$ stays within 1/10 of a unit L as soon as x stays within some distance δ of x_0.

But that in itself is no enough, because as x continues on its course towards to x_0, what is the prevent $f(x)$ jittering about within the interval $\left(L-\frac{1}{10}, L+\frac{1}{10}\right)$, without tending towards L?

We can be told that the error can be no more than 1/100 or 1/10000 or 1/100000. Each time, we find a new δ-interval about x_0 so that keeping x within that interval satisfies the new error tolerance. Each time the possibility exists that $f(x)$ jitters away from L at some stage.



By proving that for every error tolerance ϵ that can be produced, we can find, calculate or conjure a matching distance δ that keeps x "close enough" to x_0 to keep $f(x)$ within that tolerance of L.

This leads us to the precise definition of a limit.

Definition. *Limit of a Function*

Let $f(x)$ be defined on an open interval about x_0, except possibly at x_0 itself. We say that the limit of $f(x)$ as x approaches x_0 is the number L, and write

$$\lim_{x \to x_0} f(x) = L,$$

If, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all x,

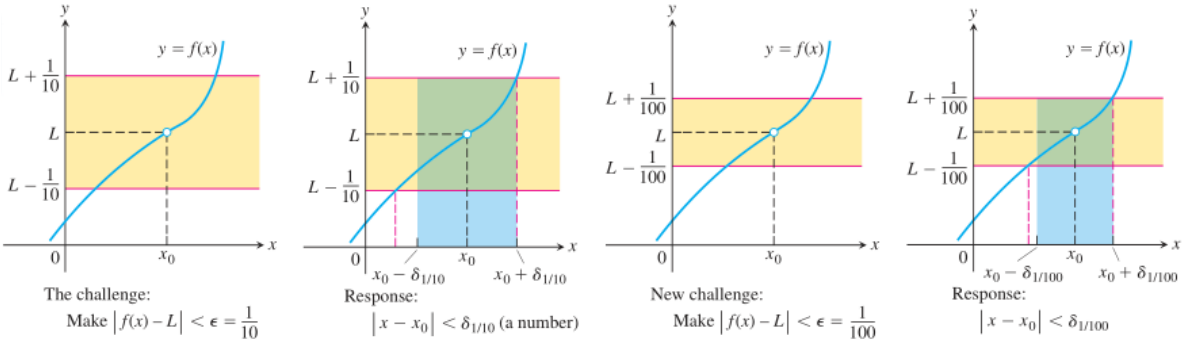
$$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

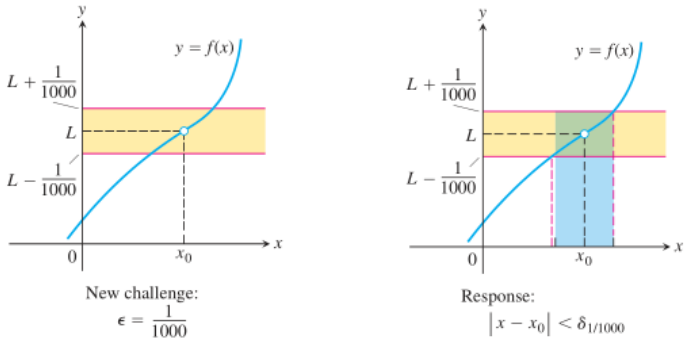
$\forall \varepsilon > 0, \exists \delta > 0$ s.t. for all x
$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$
$\lim_{x \to x_0} f(x) = L$

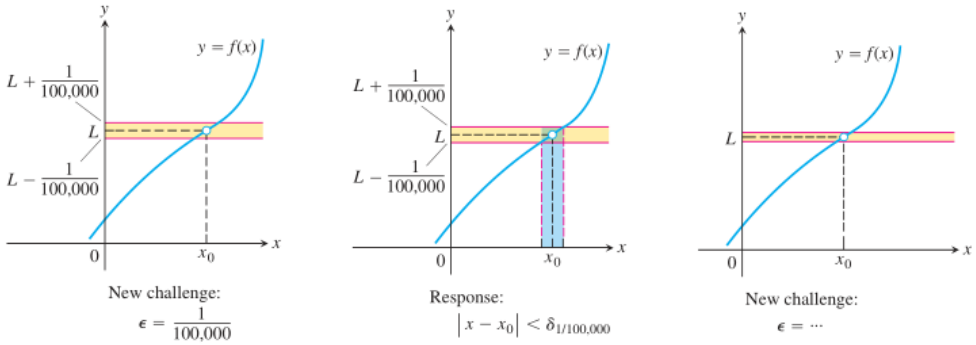
Example. *Testing the Definition*

The formal definition of limit does not tell how to find the limit of a function, but it enables us to verify that a suspected limit is correct. The following examples show how the definition can be used to verify limit statements for specific functions.

However, the real purpose of the definition is not to do calculations like this, but rather to prove general theorems so that the calculation of specific limits can be simplified.







$\lim_{x \to 1} 5x - 3 = 2$
$\forall \varepsilon > 0$, $\exists \delta = \varepsilon/10$ s.t.
if $0 < |x - 1| < \delta$, then $|5x - 3 - 2| = 5|x - 1| < 5(\varepsilon/10) < \varepsilon/2 < \varepsilon.$

Example. *Testing the Definition*

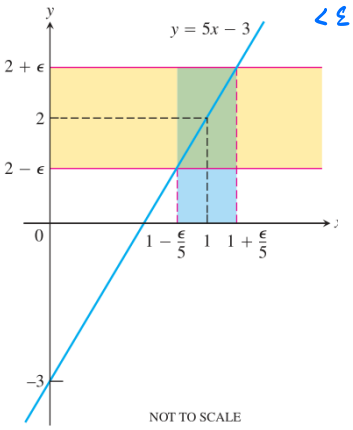
Show that $\lim_{x\to 1}(5x-3)=2$.

Solution. Set $x_0=1, f(x)=5x-3$, and $L=2$. For any given $\epsilon>0$, we have to find suitable $\delta>0$ so that if $x\neq 1$ and x is within distance δ of $x_0=1$, that is, whenever

$$0<|x-1|<\delta,$$

it is true that $f(x)$ is within distance ϵ of $L=2$, so

$$|f(x)-2|<\epsilon.$$



We find δ by working backward from the ϵ-inequality:

$$|(5x-3)-2|=|5x-5|<\epsilon$$
$$5|x-1|<\epsilon$$
$$|x-1|<\frac{\epsilon}{5}.$$

Thus, we can take $\delta=\frac{\epsilon}{5}$.

If $0<|x-1|<\delta=\epsilon/5$, then

$$|(5x-3)-2|=5|x-1|<5\left(\frac{\epsilon}{5}\right)=\epsilon,$$

which proves that $\lim_{x\to 1}5x-3=2$.

Note: The value of $\delta=\epsilon/5$ is not the only value that will make $0<|x-1|<\delta$ imply $|5x-5|<\epsilon$. Any smaller positive δ will do as well. The definition does not ask for a "best" positive δ, just one that will work.

How to Find Algebraically a δ for a Given f, L, x_0, and $\epsilon>0$.

The process of finding a $\delta>0$ such that for all x

$$0<|x-x_0|<\delta \Rightarrow |f(x)-L|<\epsilon$$

can be accomplished into two steps.

1. ***Find suitable δ.*** Solve the inequality $|f(x)-L|<\epsilon$ to find a condition of the form $|x-a|<\delta$, where δ depends on ϵ.

2. ***Write a proof.*** For any $\epsilon>0$, assume $0<|x-a|<\delta$ and use the δ found in (1), to prove $|f(x)-L|<\epsilon$.

$AB = I_n = BA$

identity

$f \circ f^{-1}$

Example. Prove the following results, which were presented graphically.

(a) $\lim_{x \to c} x = c$,

(b) $\lim_{x \to c} k = k$, k is a constant.

$\lim_{x \to 1} 4 = 4$

$\lim_{x \to c} x = c$

① Find suitable δ

$|x - c| < \varepsilon$

$0 < |x-c| < \delta$

$\delta = \varepsilon$.

② $\forall \varepsilon > 0$, $\exists \delta = \varepsilon > 0$, s.t.
if $0 < |x-c| < \delta$, then
$|x-c| < \delta < \varepsilon$.

$\Rightarrow \lim_{x \to c} x = c$.

(b) $\lim_{x \to c} k = k$, k = constant

① Find suitable δ

$|f(x) - L| = |k - k| = 0 < \varepsilon$

δ = any positive number.

② for $\forall \varepsilon > 0$, $\exists \delta = c > 0$ s.t.

$0 < |x-c| < \delta$

$\Rightarrow |k-k| = 0 < \delta = \varepsilon$.

Example. Find the limit $\lim_{x \to 5} \sqrt{x-1} = 2$, find a $\delta > 0$ that works for ε. That is, find a $\delta > 0$ such that $|\sqrt{x-1} - 2| < \varepsilon$ whenever $0 < |x-5| < \delta$.

Show: $\lim_{x \to 5} \sqrt{x-1} = 2$

$0 < |x-5| < \delta \Rightarrow |f(x) - 2| < \varepsilon$

① Find δ

$|f(x) - 2| = |\sqrt{x-1} - 2|$

$= \left| (\sqrt{x-1} - 2)\left(\frac{\sqrt{x-1}+2}{\sqrt{x-1}+2}\right) \right|$

$= \left| \frac{x-1-4}{\sqrt{x-1}+2} \right|$

$= \frac{|x-5|}{|\sqrt{x-1}+2|}$

Let $\frac{1}{|\sqrt{x-1}+2|} < C$, C = constant.

$|f(x) - 2| = \frac{|x-5|}{|\sqrt{x-1}+2|} < C|x-5| < \varepsilon$

$\Rightarrow |x-5| < \varepsilon/C$, $\delta = \varepsilon/C$, find C.

Assume $\delta = 1$

$|x-5| < 1$

$\Rightarrow -1 < x-5 < 1$

$\Rightarrow 4 < x < 6$

$\left| \frac{1}{\sqrt{x-1}+2} \right|$

$\frac{1}{\sqrt{3}+2}$, $\frac{1}{\sqrt{5}+2}$

$3 < x-1 < 5$

$\sqrt{3} < \sqrt{x-1} < \sqrt{5}$

$2+\sqrt{3} < 2+\sqrt{x-1} < 2+\sqrt{5}$

$\frac{1}{2+\sqrt{5}} < \frac{1}{\sqrt{x-1}+2} < \frac{1}{2+\sqrt{3}}$

$\left| \frac{1}{\sqrt{x-1}+2} \right| < \frac{1}{2+\sqrt{3}}$

$C = \frac{1}{2+\sqrt{3}}$

$\delta = \min\{1, (2+\sqrt{3})\varepsilon\}$.

② write down proof

$\forall \varepsilon > 0$, $\exists \delta = \min\{1, (2+\sqrt{3})\varepsilon\} > 0$, s.t.
if $0 < |x-5| < \delta$, then

$|f(x) - 2| = |\sqrt{x-1} - 2|$

$= \left| \frac{x-5}{\sqrt{x-1}+2} \right|$

$< |x-5| \left(\frac{1}{|\sqrt{x-1}+2|} \right)$

$< \frac{1}{2+\sqrt{3}} (\delta)$

$= \frac{1}{2+\sqrt{3}} ((2+\sqrt{3})\varepsilon)$

$< \varepsilon$.

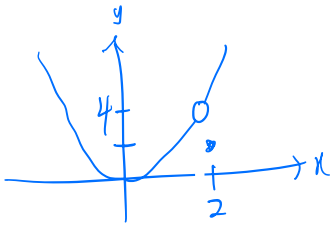
$\lim_{x \to 5} \sqrt{x-1} = 2$ #

$\lim_{x\to 1} \frac{1}{x} = 1$

Example. *Find Delta Algebraically*

Prove that $\lim_{x\to 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2 \end{cases}.$$



Solution. Our task is to show that given $\epsilon > 0$ there exists a $\delta > 0$ such that for all x

$$0 < |x-2| < \delta \Rightarrow |f(x) - 4| < \epsilon.$$

Solution. $\lim_{x\to 2} x^2 = 4$ if $0<|x-2|<\delta \Rightarrow |x^2-4|<\varepsilon$.

① Find δ

$|f(x)-4| = |x^2-4| = |(x-2)(x+2)|$

$= |x-2||x+2| < \varepsilon$

$\Rightarrow |x-2| < \frac{\varepsilon}{|x+2|}$ ✗

existing $\delta =$ ____

$|x-2|<\delta$

$\delta = 1$

$|x-2|<1$

$\Rightarrow -1 < x-2 < 1$

$\Rightarrow 3 < x+2 < 5$

$\Rightarrow |x+2| < 5$

$\Rightarrow |x+2||x-2| < \varepsilon$.

$\Rightarrow |x+2||x-2| < 5|x-2| < \varepsilon$

$\Rightarrow |x-2| < \varepsilon/5$

$\delta = \min\{1, \varepsilon/5\}$

ε, δ = numbers ! not variables.

② write proof $\delta = \min\{1, \varepsilon/5\}$

$\forall \varepsilon > 0$, $\exists \delta > 0$, s.t. all x

if $0 < |x-2| < \delta$, then

$|x^2-4| = |x+2||x-2|$

$< 5(\delta) < 5(\varepsilon/5) < \varepsilon$.

$\Rightarrow \lim_{x\to 2} x^2 = 4$.

Example. Using (ϵ, δ)-definition to prove $\lim_{x\to 3} x^2 = 9$.

Precise def.

Solution. $\lim_{x\to 3} x^2 = 9$

$0 < |x-3| < \delta \Rightarrow |x^2-9| < \varepsilon$

① Find δ

$|f(x)-9| = |x^2-9| = |(x-3)(x+3)|$ → find c

Let $|x+3| < c$, c = constant.

$|x^2-9| < c|x-3| < \varepsilon$

$\Rightarrow |x-3| < \varepsilon/c$ → $\delta = \varepsilon/c$, $c = ?$

Assume $\delta \le 1$

$|x-3|<1 \Rightarrow -1 < x-3 < 1$

$\Rightarrow 2 < x < 4$

$\Rightarrow 5 < x+3 < 7$

$\Rightarrow |x+3| < 7$

Take $c = 7$.

$\Rightarrow \delta = \min\{1, \varepsilon/7\}$

② write the proof

$\forall \varepsilon > 0$, $\exists \delta = \min\{1, \varepsilon/7\} > 0$, such that

if $0 < |x-3| < \delta$, then

$|f(x)-9| = |x^2-9|$

$= |(x-3)(x+3)|$

$= |x-3||x+3|$

$< 7(\delta)$

$< 7(\varepsilon/7)$

$< \varepsilon$.

$\Rightarrow \lim_{x\to 3} x^2 = 9$ ∎

$\lim_{x\to 1} \frac{1}{x} = 1$

$0 < |x-1| < \delta \Rightarrow |f(x)-1| < \varepsilon$.

① Find δ

$|f(x)-1| = \left|\frac{1}{x}-1\right|$

$= \left|\frac{1-x}{x}\right|$

$= \left|\frac{x-1}{x}\right|$

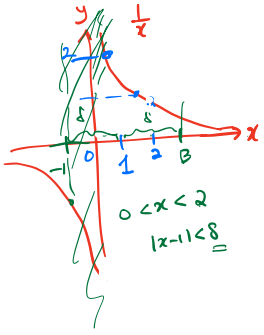
$= \left|\frac{1}{x}\right| |x-1|$

Assume $\left|\frac{1}{x}\right| < c$, c = constant.

$\left|\frac{1}{x}-1\right| < c|x-1| < \varepsilon$.

$\Rightarrow |x-1| < \varepsilon/c$

$\delta = \varepsilon/c$. Find c.



Assume $|x-1| < 1/2$

$\Rightarrow -1/2 < x-1 < 1/2$

$\Rightarrow 1/2 < x < 3/2$

$\Rightarrow 2/3 < \frac{1}{x} < 2$.

$\Rightarrow \left|\frac{1}{x}\right| < 2$

take $c = 2$.

$\delta = \min\{1/2, \varepsilon/2\}$.

② write proof

$\forall \varepsilon > 0$, $\exists \delta = \min\{1/2, \varepsilon/2\} > 0$, s.t.

if $0 < |x-1| < \delta$, then

$|f(x)-1| = \left|\frac{1}{x}-1\right|$

$= \left|\frac{x-1}{x}\right|$

$= \frac{1}{|x|}|x-1|$

$< 2\delta$

$< 2(\varepsilon/2)$

$< \varepsilon$.

$\Rightarrow \lim_{x\to 1} \frac{1}{x} = 1$.

$\lim_{x\to 1} x^2+x+1 = 3$

$0 < |x-1| < \delta \Rightarrow |f(x)-3| < \varepsilon$

① Find δ

$|x^2+x+1-3| = |x^2+x-2|$

$= |(x+2)(x-1)|$

Let $|x+2| < c$, c is constant.

Then $|x^2+x+1-3| = |x+2||x-1|$

$< c|x-1| < \varepsilon$

$\Rightarrow |x-1| < \varepsilon/c$.

$\delta = \varepsilon/c$.

Assume $|x-1| < 1$.

$\Rightarrow -1 < x-1 < 1$

$\Rightarrow 0 < x < 2$

$\Rightarrow 2 < x+2 < 4$

$\Rightarrow |x+2| < 4$. ($c=4$).

$\delta = \min\{1, \varepsilon/4\}$.

② write proof

$\forall \varepsilon > 0$, $\exists \delta = \min\{1, \varepsilon/4\} > 0$, s.t.

if $0 < |x-1| < \delta$, then

$|f(x)-3| = |x^2+x+1-3|$

$= |x^2+x-2|$

$= |x+2||x-1|$

$< 4\delta$

$< 4(\varepsilon/4) < \varepsilon$.

$\Rightarrow \lim_{x\to 1} x^2+x+1 = 3$

One-Side Limits and Limits at Infinity

We next extend the limit concept to *one-sided limits*, which are limits as x approaches the number x_0 from the left-hand side (where $x < x_0$) or the right-hand side ($x > x_0$) only.

One-Side Limits

To have a limit L as x approaches c, a function f must be defined on *both sides* of c and its values $f(x)$ must approach L as x approaches c from either side. Because of this, ordinary limits are called **two-sided**.

If $f(x)$ is defined on an interval (c, b), where $c < b$, and approaches arbitrarily close to L as x approaches c from within that interval, then f has **right-hand limit** L at c. We write

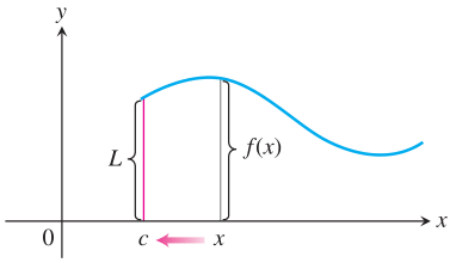
$$\lim_{x \to c^+} f(x) = L.$$

The symbol "$x \to c^+$" means that we consider only values of x greater than c.

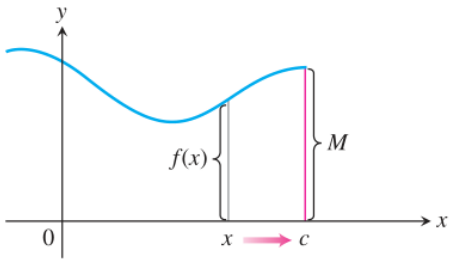
If $f(x)$ is defined on an interval (a, c), where $a < c$, and approaches arbitrarily close to L as x approaches c from within that interval, then f has **left-hand limit** L at c. We write

$$\lim_{x \to c^-} f(x) = L.$$

The symbol "$x \to c^-$" means that we consider only values of x less than c.



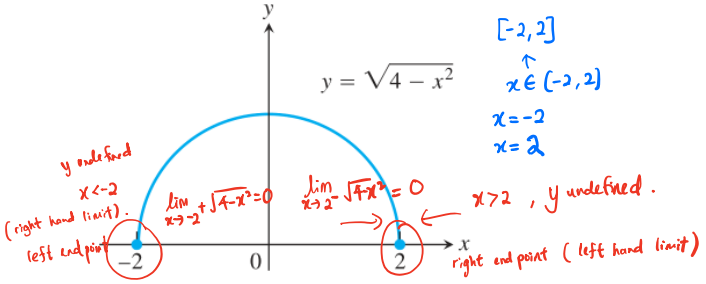
(a) $\lim_{x \to c^+} f(x) = L$



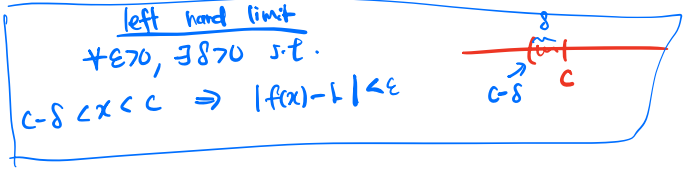
(b) $\lim_{x \to c^-} f(x) = M$

Example. *One-Sided Limits for a Semicircle*

The domain of $f(x) = \sqrt{4 - x^2}$ is $[-2,2]$; its graph is the semicircle in the following figure.



We have $\lim_{x \to -2^+} \sqrt{4 - x^2} = 0$ and $\lim_{x \to 2^-} \sqrt{4 - x^2} = 0$. The function does not have a left-hand limit at $x = -2$ or a right-hand limit at $x = 2$. It does not have ordinary two-sided limits at either -2 or 2.



Definitions. *Right-Hand, Left-Hand Limits*

We say that $f(x)$ has **right-hand limit L at c**, and write

$$\lim_{x\to c^+} f(x) = L$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$c < x < c + \delta \;\Rightarrow\; |f(x) - L| < \epsilon.$$

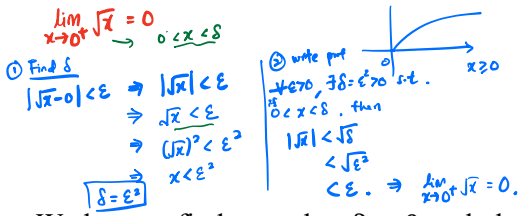
We say that $f(x)$ has **left-hand limit L at c**, and write

$$\lim_{x\to c^-} f(x) = L$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$c - \delta < x < c \;\Rightarrow\; |f(x) - L| < \epsilon.$$

Example.
Using (ϵ, δ)-definition to prove $\lim_{x\to 0^+} \sqrt{x} = 0$.

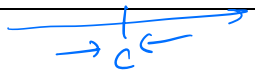


Solution.

1. *Guessing a value for* δ. Let $\epsilon > 0$ be given. We have to find a number $\delta > 0$ such that
$$|\sqrt{x} - 0| < \epsilon \text{ whenever } 0 < x < \delta$$
That is, $|\sqrt{x}| < \epsilon$ whenever $0 < x < \delta$, or squaring both sides of the inequality $\sqrt{x} < \epsilon$, we get $x < \epsilon^2$, whenever $0 < x < \delta$.
This suggests that we should choose $\delta = \epsilon^2$.

2. *Showing that this* δ *works.* Given $\epsilon > 0$, let $\delta = \epsilon^2$. If $0 < x < \delta$, then
$$\sqrt{x} < \sqrt{\delta} = \sqrt{\epsilon^2} = \epsilon.$$
So, $$|\sqrt{x} - 0| < \epsilon.$$
According to the definition, this shows that $\lim_{x\to 0^+} \sqrt{x} = 0$.

Theorem.

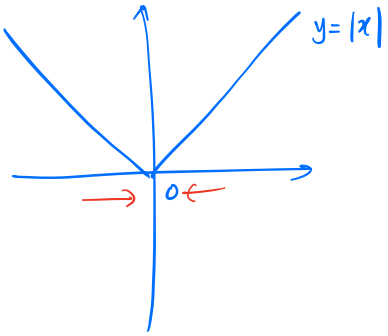


A function $f(x)$ has a limit as x approaches c if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x\to c} f(x) = L \Leftrightarrow \lim_{x\to c^-} f(x) = L \text{ and } \lim_{x\to c^+} f(x) = L.$$

$\lim_{x\to c} f(x) = L \iff \lim_{x\to c^-} f(x) = L$ and $\lim_{x\to c^+} f(x) = L$.

$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$



Example. Show that $\lim_{x \to 0} |x| = 0$.

Solution

$\lim_{x \to 0^+} |x| = \lim_{x \to 0^+} x = 0$

$\lim_{x \to 0^-} |x| = \lim_{x \to 0^-} (-x) = 0$

$\lim_{x \to 0^+} |x| = \lim_{x \to 0^-} |x| = 0 \Rightarrow \lim_{x \to 0} |x| = 0.$

Example. Show that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

Solution

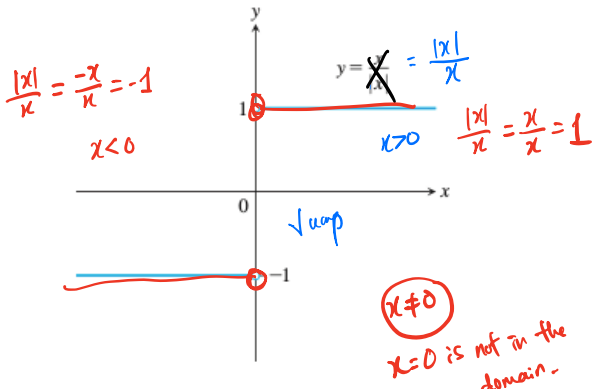


FIGURE 2.24 Different right-hand and left-hand limits at the origin.

Show: $\lim_{x \to 0} \frac{|x|}{x} \neq$

$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} = \lim_{x \to 0^+} 1 = 1$

$\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} = \lim_{x \to 0^-} (-1) = -1$ exist

$x < 0$

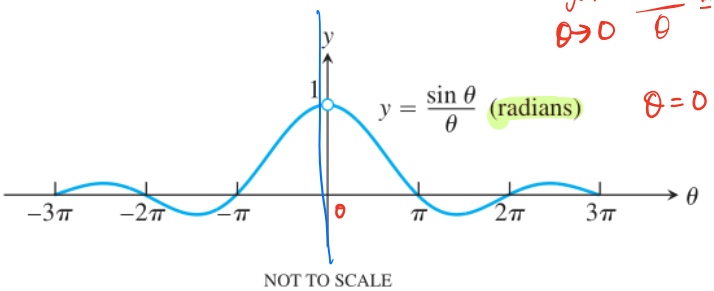
But $\lim_{x \to 0^-} \frac{|x|}{x} \neq \lim_{x \to 0^+} \frac{|x|}{x}$

$\Rightarrow \lim_{x \to 0} \frac{|x|}{x}$ doesn't exist.

Limits Involving $\frac{\sin \theta}{\theta}$

$f(\theta) = \frac{\sin \theta}{\theta}$ even

$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1$

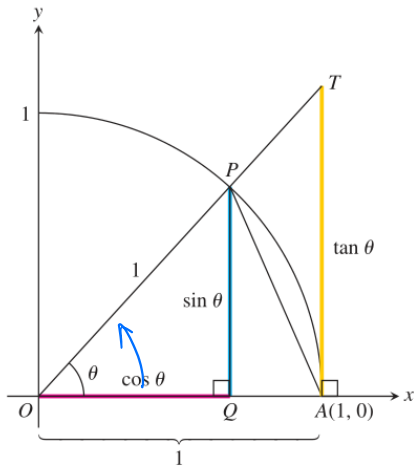


θ = 0 not in domain.

According to the figure above, we can see that the limit of $\sin \theta / \theta$ as $\theta \to 0$ is 1. We can confirm it algebraically using Sandwich Theorem.

Theorem. $\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1$, ($\theta$ in radians)

Proof.



We note that $Area\ OAP < Area\ section\ OAP < Area\ OAT$

Area $OAP = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}(1)(\sin\theta) = \frac{1}{2}\sin\theta$

Area section $OAP = \frac{1}{2}r^2\theta = \frac{1}{2}(1)^2\theta = \frac{\theta}{2}$

Area $OAT = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}\tan\theta$

Thus, $\frac{1}{2}\sin\theta < \frac{\theta}{2} < \frac{1}{2}\tan\theta$.

Divide all three terms by the number $\frac{1}{2}\sin\theta$, and we note that $0 < \theta < \frac{\pi}{2}$, hence, $\sin\theta > 0$.

$$\Rightarrow 1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}.$$

Taking reciprocals reverses,

$$\Rightarrow 1 > \frac{\sin\theta}{\theta} > \cos\theta.$$

Since $\lim_{\theta\to 0^+}\cos\theta = 1$, by Sandwich Theorem, $\lim_{\theta\to 0^+}\frac{\sin\theta}{\theta} = 1$

Note $\frac{\sin\theta}{\theta}$ is an even function, with a graph symmetric about y-axis.

Therefore, $\lim_{\theta\to 0^-}\frac{\sin\theta}{\theta} = 1 = \lim_{\theta\to 0^+}\frac{\sin\theta}{\theta}$. So, $\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$.

Example. Show the following

(a) $\lim_{h\to 0}\frac{\cos h - 1}{h} = 0$

(b) $\lim_{x\to 0}\frac{\sin(2x)}{(5x)} = \frac{2}{5}$

Solution.

(a) Using half-angle formula, $\cos h = 1 - 2\sin^2\frac{h}{2}$, we calculate

$$\begin{aligned}\lim_{h\to 0}\frac{\cos h - 1}{h} &= \lim_{h\to 0}\frac{-2\sin^2\frac{h}{2}}{h}\\ &= \lim_{h\to 0}\left(\frac{\sin\frac{h}{2}}{\frac{h}{2}}\right)(-\sin h)\\ &= (1)(0) = 0\end{aligned}$$

$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$

(b) $\lim_{x\to 0}\frac{\sin 2x}{5x}=\frac{2}{5}\lim_{x\to 0}\frac{\sin 2x}{2x}=\frac{2}{5}(1)=\frac{2}{5}.$

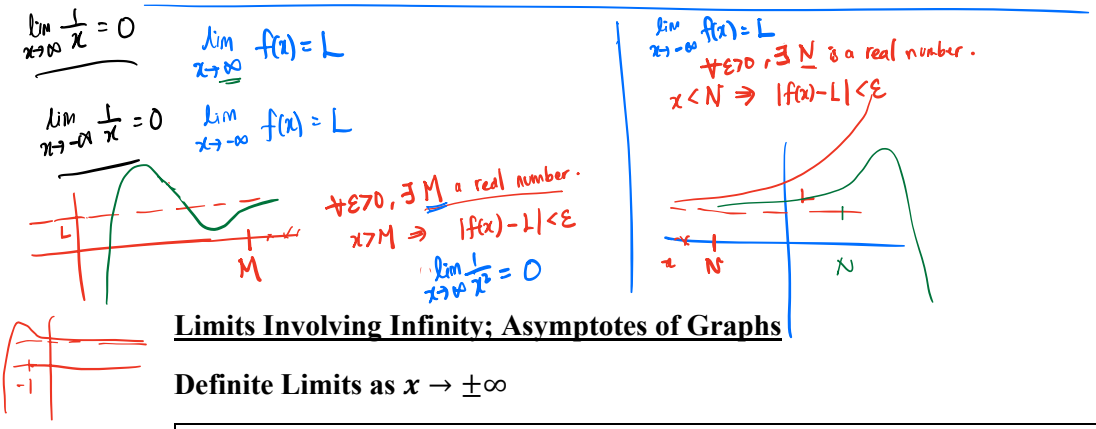
adjust $\frac{1}{2}$ $(x\to 0)\left(\frac{x}{x}\to 1\right)$ $x\neq 0$.

Exercises:

1. $\lim_{x\to 0}\frac{\sin(3x)}{\sin(2x)}=\lim_{x\to 0}\left(\frac{\sin(3x)}{3x}\right)\cdot\left(\frac{2x}{\sin(2x)}\right)\cdot\frac{3}{2}$

$=3/2.$

2. $\lim_{\theta\to 0}\frac{\sin A\theta}{\sin B\theta}=\lim_{\theta\to 0}\frac{\sin A\theta}{A\theta}\cdot\frac{B\theta}{\sin B\theta}\cdot\frac{A}{B}=\frac{A}{B}.$



Limits Involving Infinity; Asymptotes of Graphs

Definite Limits as $x \to \pm\infty$

Definition. *Limit as x approaches ∞ or −∞*

1. We say that $f(x)$ has **the limit *L* as *x* approaches infinity** and write
$$\lim_{x\to\infty} f(x)=L$$
If, for every number $\epsilon>0$, there exists a corresponding number M such that for all x
$$x>M \;\Rightarrow\; |f(x)-L|<\epsilon.$$

2. We say that $f(x)$ has **the limit *L* as *x* approaches minus infinity** and write
$$\lim_{x\to-\infty} f(x)=L$$
If, for every number $\epsilon>0$, there exists a corresponding number N such that for all x
$$x<N \;\Rightarrow\; |f(x)-L|<\epsilon.$$

Intuitively, $\lim_{x\to\infty} f(x)=L$ if, as x moves increasingly far from the origin in the positive direction, $f(x)$ get arbitrarily close to L.

Similarly, $\lim_{x\to-\infty} f(x)=L$ if as x moves increasingly far from the origin in the negative direction, $f(x)$ gets arbitrarily close to L.

Example. *Limits at Infinity for* $f(x)=\frac{1}{x}$

Show that

(a) $\lim_{x\to\infty}\frac{1}{x}=0$ (b) $\lim_{x\to-\infty}\frac{1}{x}=0.$

(i) $\lim_{x\to\infty}\frac{1}{x}=0 \to x>M \Rightarrow |\frac{1}{x}-0|<\varepsilon.$

① Find M

$|\frac{1}{x}-0|<\varepsilon$

$\Rightarrow |\frac{1}{x}|<\varepsilon$

$\Rightarrow \frac{1}{|x|}<\varepsilon.$

$\Rightarrow |x|>1/\varepsilon$

$\Rightarrow x<-1/\varepsilon,\ x>1/\varepsilon$

choose $M=1/\varepsilon$

② Write Proof.

$\forall\varepsilon>0$, $\exists M=1/\varepsilon$ (real number) s.t. if $x>M$, then

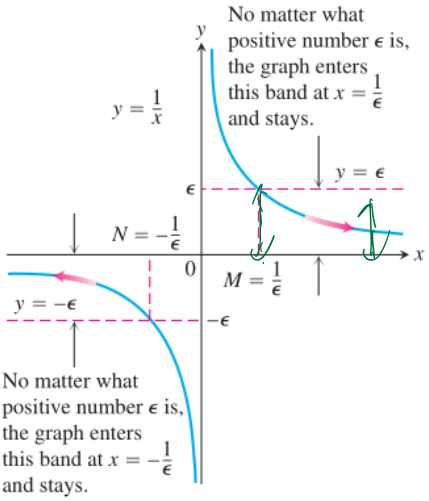
$|\frac{1}{x}|<\frac{1}{|x|}<1/M.$

$x>M$

$|x|>|M|>1/\varepsilon.$ $<1/(1/\varepsilon)$

$\frac{1}{|x|}<\varepsilon.$ $<\varepsilon.$

$\to |f(x)-0|<\varepsilon.$



Solution.

(a) Let $\epsilon > 0$ be given. We first find a number M such that for all x

$$x > M \Rightarrow \left|\frac{1}{x} - 0\right| = \left|\frac{1}{x}\right| < \epsilon.$$

This implication holds if $M = \frac{1}{\epsilon}$ or any larger positive number. This proves $\lim_{x\to\infty}\left(\frac{1}{x}\right) = 0$.

(b) Let $\epsilon > 0$ be given. We first find a number N such that for all x

$$x < N \Rightarrow \left|\frac{1}{x} - 0\right| = \left|\frac{1}{x}\right| < \epsilon.$$

This implication holds if $N = -\frac{1}{\epsilon}$ or any larger positive number. This proves $\lim_{x\to-\infty}\left(\frac{1}{x}\right) = 0$.

Theorem. *Limit Laws as $x \to \pm\infty$*
If L. M. and k, are real numbers and

$$\lim_{x\to\pm\infty} f(x) = L \text{ and } \lim_{x\to\pm\infty} g(x) = M$$

1. $\lim_{x\to\pm\infty}\big(f(x) + g(x)\big) = L + M$
2. $\lim_{x\to\pm\infty}\big(f(x) - g(x)\big) = L - M$
3. $\lim_{x\to\pm\infty}\big(f(x)\cdot g(x)\big) = LM$
4. $k\lim_{x\to\pm\infty} f(x) = kL$
5. $\lim_{x\to\pm\infty}\left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$
6. If r and s are integers with no common factors, $s \neq 0$, then

$$\lim_{x\to\pm\infty}\big(f(x)\big)^{\frac{r}{s}} = L^{\frac{r}{s}}$$

provided that $L^{\frac{r}{s}}$ is a real number. (if s is even, we assume that $L > 0$.)

Example. *Using Theorem*

(a) $\lim_{x\to\infty}\left(5 + \frac{1}{x}\right) = \lim_{x\to\infty} 5 + \lim_{x\to\infty}\frac{1}{x} = 5 + 0 = 5$

(b) $\lim_{x\to\infty}\left(\frac{\pi\sqrt{3}}{x^2}\right) = \lim_{x\to\infty}\left(\pi\sqrt{3}\cdot\frac{1}{x}\cdot\frac{1}{x}\right) = \lim_{x\to\infty}\left(\pi\sqrt{3}\right)\cdot\lim_{x\to\infty}\frac{1}{x}\cdot\lim_{x\to\infty}\frac{1}{x} = \pi\sqrt{3}(0)(0) = 0.$

Limits at Infinity of Rational Functions

To determine the limit of a rational function as $x \to \infty$, we first divide the numerator and denominator by the highest power of x in the denominator.

The result then depends on the degrees of the polynomials denominator.

Example:

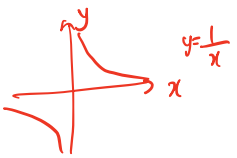
These examples illustrate what happens when the degree of the numerator is less than or equal to the degree of the denominator.

(a) $\lim_{x\to\infty} \frac{5x^2+8x-3}{3x^2+2}$

(b) $\lim_{x\to\infty} \frac{11x+2}{2x^3-1}$

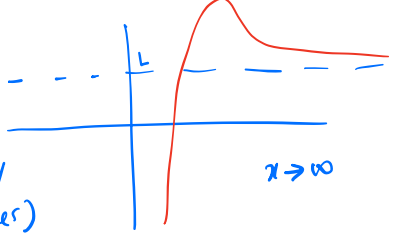
(c) $\lim_{x\to\infty} \frac{3x+7}{x^2-2}$

(d) $\lim_{x\to\infty} \frac{3x^3+7}{x^2-2}$



$x \to +\infty$, $f(x) \to 0$, $y=0$ horizontal asy.

$x \to -\infty$, $f(x) \to 0$, $y=0$ horizontal asy.



$x \to \pm\infty$, $f(x) \to L$

$y = L$ (horizontal asymptotes)

Horizontal Asymptotes

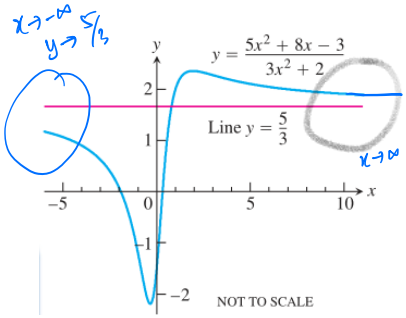
If the *distance* between the graph of a function and some fixed line approaches zero as a point on the graph *moves increasingly far from the origin*, we say that the graph approaches the line asymptotically and that the line is an *asymptote* of the graph.

Definition. *Horizontal Asymptotes*

A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x\to\infty} f(x) = b \quad \text{or} \quad \lim_{x\to-\infty} f(x) = b$$

Example. *Numerator and Denominator of Same Degree*



$x \to -\infty$, $y \to 5/3$

The curve $f(x) = \frac{5x^2+8x-3}{3x^2+2}$, we note that

$\deg(p) = 2$, $\deg(q) = 2$

$\lim_{x\to\pm\infty} 1/x = 0$, $\lim_{x\to\pm\infty} 1/x^n = 0$ $n>0$

$\lim_{x\to\pm\infty} 1/x^2 = 0$

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{5x^2+8x-3}{3x^2+2} = 5/3$

$y = 5/3$

$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{5x^2+8x-3}{3x^2+2} = \lim_{x\to-\infty} \frac{5x^2/x^2 + 8x/x^2 - 3/x^2}{3x^2/x^2 + 2/x^2}$

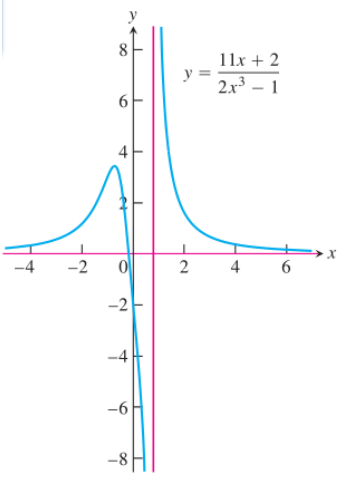
$= \lim_{x\to-\infty} \frac{5 + 8/x - 3/x^2}{3 + 2/x^2} = 5/3.$

$x \to -\infty$

$y \to 5/3$

The line $y = \frac{5}{3}$ as a horizontal asymptote on both the left and the right.

Example. *Degree of Numerator Less than Degree of Denominator*



The curve $f(x) = \frac{11x+2}{2x^3-1}$, we note that

$\deg(q) = 1$

$\deg(q) = 3$

$\lim_{x\to\infty} \frac{(11x+2)}{(2x^3+1)} = 0^+$

$\lim_{x\to-\infty} \frac{11x+2}{2x^3+1} = 0^+$ (−ve / −ve)

$y = 0$ horizontal asymptote.

The line $y = 0$ as a horizontal asymptote on both the left and the right.

Example. Find the horizontal asymptotes of the graph of

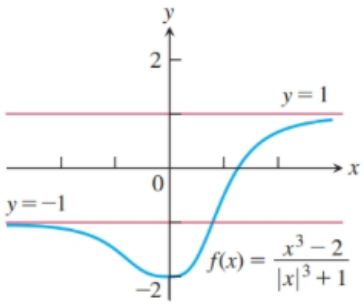


FIGURE 2.39 The graph of the function in Example 4 has two horizontal asymptotes.

The curve $f(x) = \frac{x^3-2}{|x|^3+1}$, we note that

For $x > 0$: $\lim_{x\to\infty} \frac{x^3-2}{|x|^3+1} = \lim_{x\to\infty} \frac{x^3-2}{x^3+1} = 1$

For $x < 0$: $\lim_{x\to-\infty} \frac{x^3-2}{|x|^3+1} = \lim_{x\to-\infty} \frac{x^3-2}{(-x)^3+1} = \lim_{x\to-\infty} \frac{x^3-2}{-x^3+1} = -1$

Example:

Find (a) $\lim_{x\to\infty} \sin\left(\frac{1}{x}\right)$, and (b) $\lim_{x\to\pm\infty} x \sin\left(\frac{1}{x}\right)$.

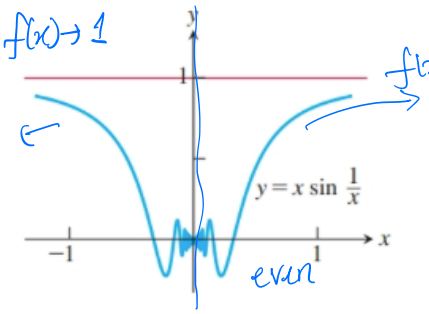


FIGURE 2.40 The line $y = 1$ is a horizontal asymptote of the function graphed here (Example 5b).

(a) $\lim_{x\to\infty} \sin\left(\frac{1}{x}\right) = \sin\left(\lim_{x\to\infty} \frac{1}{x}\right) = \sin 0 = 0$

(b) $\lim_{x\to\infty} x \sin\left(\frac{1}{x}\right) = \lim_{x\to\infty} \frac{\sin(1/x)}{1/x}$

Let $t = 1/x$. $x \to \infty$, $t \to 0^+$

$\lim_{t\to 0^+} \frac{\sin t}{t} = 1$

$\lim_{t\to 0^-} \frac{\sin t}{t} = 1$

$\Rightarrow \lim_{x\to\pm} x \sin(\frac{1}{x}) = 1$

$y = 1$ horizontal asymptote.

$f(t) = \frac{\sin t}{t}$

$f(-t) = \frac{\sin(-t)}{-t} = \frac{-\sin t}{-t} = \frac{\sin t}{t} = f(t)$

f even.

$-1 \le \sin(\frac{1}{x}) \le 1$

$-|x| \le x \sin(\frac{1}{x}) \le |x|$

$\lim_{x\to\infty} -|x| = -\infty$ (L)

$\lim_{x\to\infty} |x| = \infty$ (L)

$\lim_{\theta\to 0} \frac{\sin\theta}{\theta}$.

Example: Find $\lim_{x\to 0^+} x \left\lfloor \frac{1}{x} \right\rfloor$.

$\lfloor x \rfloor$

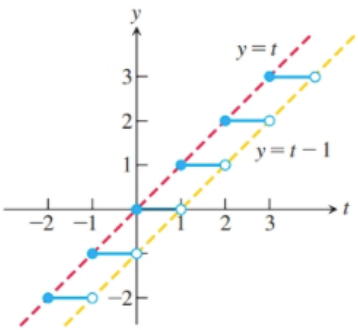


FIGURE 2.41 The graph of the greatest integer function $y = \lfloor t \rfloor$ is sandwiched between $y = t - 1$ and $y = t$.

floor integer function

$\lim_{x\to 0^+} x \left\lfloor \frac{1}{x} \right\rfloor$

$t = 1/x$ $\quad x \to 0^+, t \to +\infty$

$\lim_{t\to\infty} \frac{1}{t} \lfloor t \rfloor$

$t - 1 \le \lfloor t \rfloor \le t + 1$

$\frac{1}{t}(t-1) \le \frac{1}{t}\lfloor t \rfloor \le \frac{1}{t}(t+1)$

$1 - 1/t \le \frac{1}{t}\lfloor t \rfloor \le 1 + 1/t$

$\lim_{t\to\infty} (1 - 1/t) = 1$, $\lim_{t\to\infty} (1 + 1/t) = 1$

By Sandwich theorem, $\lim_{t\to\infty} \frac{1}{t}\lfloor t \rfloor = 1 \Rightarrow \lim_{x\to 0^+} x \left\lfloor \frac{1}{x} \right\rfloor = 1$.

The *Sandwich Theorem also holds for limits as* $(x \to \infty)$. You must be sure, though, that the function whose limit you are trying to find stays between the bounding functions for all very large x in the positive direction (if $x \to \infty$) or all very large x in the negative (if $x \to -\infty$).

$\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$

Example: Use Sandwich Theorem, find the horizontal asymptote of the curve

$$y = 2 + \frac{\sin x}{x}.$$

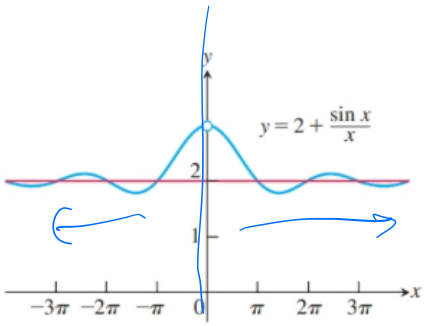


FIGURE 2.42 A curve may cross one of its asymptotes infinitely often (Example 7).

$\lim_{x\to\infty} 2 + \frac{\sin x}{x} = \left(\lim_{x\to\infty} 2\right) + \left(\lim_{x\to\infty} \frac{\sin x}{x}\right)$

$= 2 + 0 = 2$

$\lim_{x\to-\infty} 2 + \frac{\sin x}{x} = 2$

$y = 2$ horizontal asymptote.

$-1 \le \sin x \le 1$

$-\frac{1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$, $x \to \infty$, $x > 0$.

$\lim_{x\to\infty} -\frac{1}{x} = 0$, $\lim_{x\to\infty} \frac{1}{x} = 0$

By Sandwich theorem,

$\lim_{x\to\infty} \frac{\sin x}{x} = 0$

$\infty - \infty$

Example: Find $\lim_{x\to\infty} \left(x - \sqrt{x^2 + 16}\right)$.

$\lim_{x\to\infty} \left(x - \sqrt{x^2+16}\right)\left(\frac{x+\sqrt{x^2+16}}{x+\sqrt{x^2+16}}\right)$

$= \lim_{x\to\infty} \frac{x^2 - (x^2+16)}{x+\sqrt{x^2+16}}$

$= \lim_{x\to\infty} \frac{-16}{x+\sqrt{x^2+16}} = 0$

---

eg: $\lim_{x\to 0^+} 2\sqrt{x}\left(1 + \sin^2\left(\frac{5\pi}{6x}\right)\right) = ?$

$0 \le \sin^2\left(\frac{5\pi}{6x}\right) \le 1$

$1 \le 1 + \sin^2\left(\frac{5\pi}{6x}\right) \le 2$

$2\sqrt{x} \le 1 + \sin^2\left(\frac{5\pi}{6x}\right) \le 4\sqrt{x}$

$\lim_{x\to 0^+} 2\sqrt{x} = 0$

$\lim_{x\to 0^+} 4\sqrt{x} = 0$

By Sandwich thm,

$\lim_{x\to 0^+} 2\sqrt{x}\left(1 + \sin^2\left(\frac{5\pi}{6x}\right)\right) = 0.$

Oblique Asymptotes (Slanted asymptotes)

If the degree of the numerator of a rational function is one greater than the degree of the denominator, the graph has an **oblique (slanted) asymptote**. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \to \pm\infty$.

$x \to \pm\infty$ $y \to L$ ($y = L$ horizontal)

$x \to \pm\infty$ $y \to \infty$

$y \to f(x)$ ($y = f(x)$ slanted asymptote).

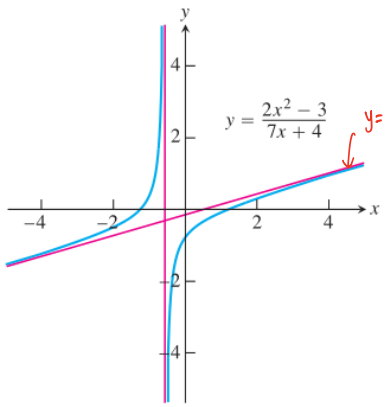
$\lim_{x\to\pm\infty} f(x) = L$ ↑ $\lim_{x\to\infty} \frac{P(x)}{q(x)} = L$ ↑ $\frac{\text{leading coe. } P}{\text{leading coe. } q}$

$\deg(p) = \deg(q)$

$\lim_{x\to\infty} \frac{P(x)}{q(x)} = \infty$ $\deg p > \deg q$.

Example. Find the oblique (slanted) asymptote for the graph of $f(x) = \frac{2x^2-3}{7x+4}$.

Solution.



By long division, we find

$$f(x) = \frac{2x^2-3}{7x+4} = \left(\frac{2}{7}x - \frac{8}{49}\right) + \frac{-115}{49(7x+4)}.$$

As $x \to \pm\infty$, the remainder, whose magnitude gives the vertical distance between the graphs of f and g, goes to zero, making the (slanted) line

$$g(x) = \frac{2}{7}x - \frac{8}{49}$$

an asymptote of the graph of f.

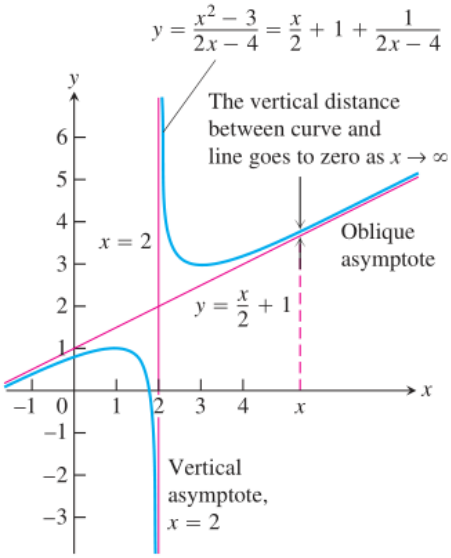
Example. *A Rational Function with Degree of Numerator Greater than Degree of Denominator*

Find the asymptotes of the graph of

$$f(x) = \frac{x^2-3}{2x-4}.$$

Solution. First, we note

$$f(x) = \frac{x^2-3}{2x-4} = \frac{x}{2} + 1 + \frac{1}{2x-4}.$$



We are interested in the behaviour as $x \to \pm\infty$ and also as $x \to 2$, where the denominator is zero.

As $x \to \pm\infty$, the remainder approaches zero, and $f(x) \to \frac{x}{2} + 1$.

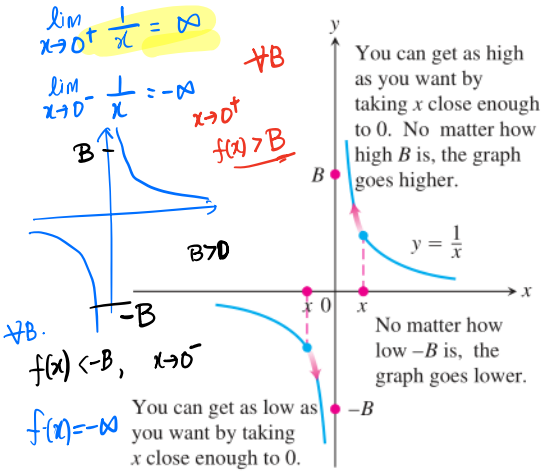
The line $y = \frac{x}{2} + 1$ as an oblique (slanted) asymptote, both to the right and to the left.

As $x \to 2^+$, $f(x) \to +\infty$ and as $x \to 2^-$, $f(x) \to -\infty$. Hence, the line $x = 2$ is a two-sided vertical asymptote.

Infinite Limits and Vertical Asymptotes

Example.

We consider $f(x) = \frac{1}{x}$. As $x \to 0^+$, the values of f grow without bound, eventually reaching and surpassing every positive real number. That is, given any *positive real number* B, however large, the values of f become larger still. Thus, f has no limit as $x \to 0^+$.



We write $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{x} = \infty$.

Note that, we are not saying that the limit exists. Nor are we saying that there is a real number ∞, for there is no such number.

We say, $\lim_{x\to 0^+} \frac{1}{x}$ does not exist because $1/x$ becomes arbitrarily large and positive as $x \to 0^+$.

As $x \to 0^-$, the values of $f(x) = 1/x$ become arbitrarily large and negative. Given any negative real number $-B$, the value of f eventually lie below $-B$.

We write $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{x} = -\infty$.

We are describing the behavior of a function whose limit as $x \to 0^-$ *does not exist because its values become arbitrarily large and negative.*

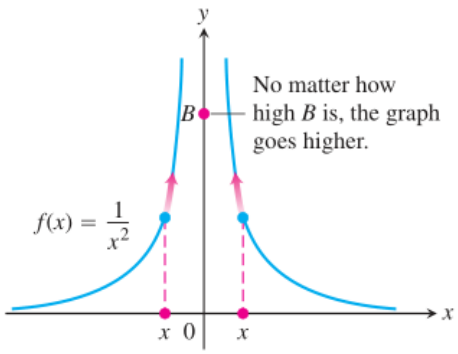
Example. *Two-sided Infinite Limits*

Discuss the behavior of

(a) $f(x) = \frac{1}{x^2}$ near $x = 0$.

(b) $g(x) = \frac{1}{(x+3)^2}$ near $x = -3$.

Solution.



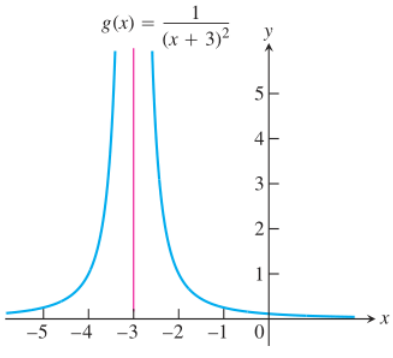
(a)

(a) As $x \to 0$ from either side, the values of $1/x^2$ are positive and become arbitrarily large.

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1}{x^2} = \infty.$$

*The function $y = 1/x$ shows no consistent behavior as $x \to 0$. We have $\frac{1}{x} \to \infty$ $(x \to 0^+)$, $\frac{1}{x} \to -\infty$ $(x \to 0^-)$. We say $\lim_{x\to 0} 1/x$ does not exist.

But $y = 1/x^2$ is different. Its values approach infinity as $x \to 0$ from either side. So, we can say $\lim_{x\to 0} 1/x^2 = \infty$.



(b)

(b) We note that $g(x) = 1/(x+3)^2$ is the graph of $f(x) = 1/x^2$ shifted 3 units to the left. Therefore, g behaves near -3 exactly the way f behaves near 0.

$$\lim_{x\to -3} g(x) = \lim_{x\to -3} \frac{1}{(x+3)^2} = \infty.$$

Example: Find $\lim_{x\to 1^+} \frac{1}{x-1}$ and $\lim_{x\to 1^-} \frac{1}{x-1}$.

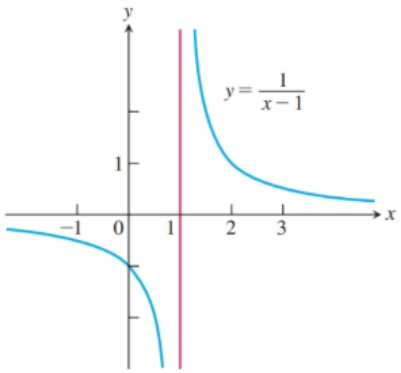
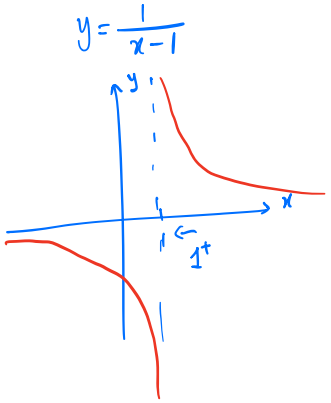


FIGURE 2.45 Near $x = 1$, the function $y = 1/(x - 1)$ behaves the way the function $y = 1/x$ behaves near $x = 0$. Its graph is the graph of $y = 1/x$ shifted 1 unit to the right (Example 10).



$\lim_{x\to 1^+} \frac{1}{x-1} = \infty$

$\lim_{x\to 1^-} \frac{1}{x-1} = -\infty$

$\lim_{x\to 1} \frac{1}{x-1}$ ✗ $= \infty$

Example: Discuss the behavior of

$$f(x) = \frac{1}{x^2} \quad \text{as} \quad x \to 0$$

↳ $\lim_{x\to 0} \frac{1}{x^2} = \infty$.

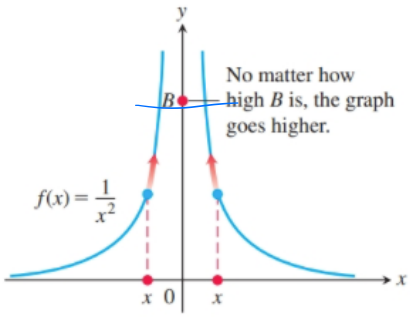


FIGURE 2.46 The graph of $f(x)$ in Example 11 approaches infinity as $x \to 0$.

Example: These examples illustrate that rational functions can behave in various ways near zeros of the denominator.

(a) $\lim_{x\to 2} \frac{(x-2)^2}{x^2-4}$ ($0/0$) $= \lim_{x\to 2} \frac{(x-2)^2}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x-2}{x+2} = \frac{0}{4} = 0$.

(b) $\lim_{x\to 2} \frac{x-2}{x^2-4}$ ($0/0$) $= \lim_{x\to 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x\to 2} \frac{1}{x+2} = \frac{1}{4}$.

(c) $\lim_{x\to 2^+} \frac{x-3}{x^2-4}$ ($-1/0$) $= \lim_{x\to 2^+} \frac{(x-3)}{(x-2)(x+2)} = -\infty$

(-1 (−ve); 0^+ (+ve); (+ve))

(d) $\lim_{x\to 2^-} \frac{x-3}{x^2-4}$ $= \lim_{x\to 2^-} \frac{(x-3)}{(x-2)(x+2)} = +\infty$

(e) $\lim_{x\to 2} \frac{x-3}{x^2-4}$ ∄ (no conclusion)

(f) $\lim_{x\to 2} \frac{2-x}{(x-2)^3} = \lim_{x\to 2} \frac{-(x-2)}{(x-2)^3} = \lim_{x\to 2} \frac{(-1)}{(x-2)^2} = -\infty$

Example: Find $\lim_{x\to-\infty} \frac{2x^5-6x^4+1}{3x^2+x-7}$.

$$= \lim_{x\to-\infty} \frac{2x^3-6x^2+1/x^2}{3+1/x-7/x^2}$$

$$= \lim_{x\to-\infty} \frac{2x^3(x-3)+1/x^2}{3+1/x-7/x^2}$$

$$= -\infty$$

$\forall \varepsilon > 0, \exists \delta > 0$ s.t.
if $0 < |x-a| < \delta$, then $|f(x)-L| < \varepsilon$.

$\lim_{x\to a} f(x) = L$

$\lim_{x\to a} f(x) = \infty$

$\forall B, \exists \delta > 0$ s.t.

$0 < |x-a| < \delta \Rightarrow f(x) > B$

$\lim_{x\to a} f(x) = -\infty$

$\forall B > 0, \exists \delta > 0$

$0 < |x-a| < \delta \Rightarrow f(x) < -B$.

$\lim_{x\to a} f(x) = \infty$ / $\lim_{x\to a} f(x) = -\infty$

Precise Definitions of Infinite Limits.

Definitions. *Infinity, Negative Infinity as Limits*

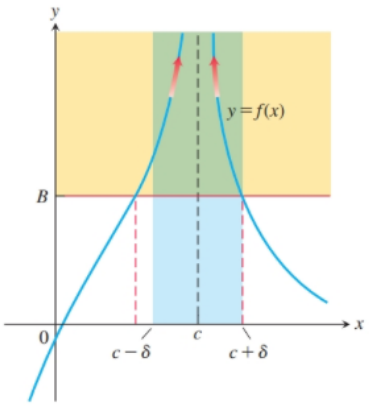


FIGURE 2.47 For $c - \delta < x < c + \delta$, the graph of $f(x)$ lies above the line $y = B$.

1. We say that $f(x)$ **approaches infinity as** x **approaches** x_0, and write

$$\lim_{x\to x_0} f(x) = \infty,$$

If for every positive real number B there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - c| < \delta \;\Rightarrow\; f(x) > B.$$

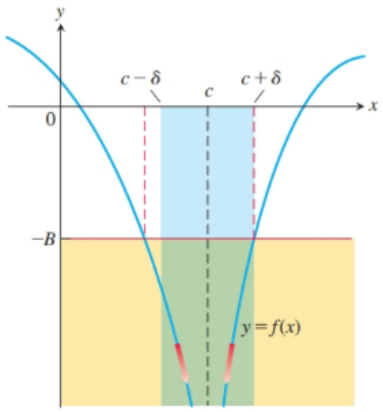


FIGURE 2.48 For $c - \delta < x < c + \delta$, the graph of $f(x)$ lies below the line $y = -B$.

2. We say that $f(x)$ **approaches negative infinity as** x **approaches** x_0, and write

$$\lim_{x\to x_0} f(x) = -\infty,$$

If for every negative real number $-B$ there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - c| < \delta \;\Rightarrow\; f(x) < -B.$$

$\lim_{x\to\infty} \frac{1}{x^2} = 0$ ↘

$\lim_{x\to\infty} f(x) = L$
$\exists M$, $\forall \varepsilon > 0$
$x > M \Rightarrow |f(x) - 0| < \varepsilon$

Example. *Using Definition of Infinite Limits*

Prove that $\lim_{x\to 0} \frac{1}{x^2} = \infty$.

Solution. Given $B > 0$, we want to find $\delta > 0$ such that

$$0 < |x - 0| < \delta \;\Rightarrow\; \frac{1}{x^2} > B.$$

Now, $\frac{1}{x^2} > B \Leftrightarrow x^2 < \frac{1}{B}$, or equivalently,

$$|x| < \frac{1}{\sqrt{B}}.$$

Thus, we can choose $\delta = 1/\sqrt{B}$ (or any smaller positive number), we see that

$$|x| < \delta \;\Rightarrow\; \frac{1}{x^2} > \frac{1}{\delta^2} \geq B.$$

$\lim_{x\to 0} \frac{1}{x^2} = \infty$
$\forall B > 0$, $\exists \delta > 0$
① Find δ
$\frac{1}{x^2} > B$
$\frac{1}{\sqrt{x^2}} > \sqrt{B}$
$\frac{1}{|x|} > \sqrt{B}$
$|x| < \frac{1}{\sqrt{B}}$
$\delta = \frac{1}{\sqrt{B}}$

$\alpha\ 0 < |x-0| < \delta \Rightarrow \frac{1}{x^2} > B$.
$|x| < \delta$

② write proof.
$\forall B > 0$, $\exists \delta = \frac{1}{\sqrt{B}} > 0$
s.t.
if $\alpha\ |x-0| < \delta$, then
$\frac{1}{x^2} > \frac{1}{\delta^2}$
$\frac{1}{|x|} > \frac{1}{\delta}$
$\Rightarrow \frac{1}{x^2} > \frac{1}{(\sqrt{B})^2}$
$> (\sqrt{B})^2 > B$.
$\Rightarrow \frac{1}{x^2} > B$
$\lim_{x\to 0} \frac{1}{x^2} = \infty$

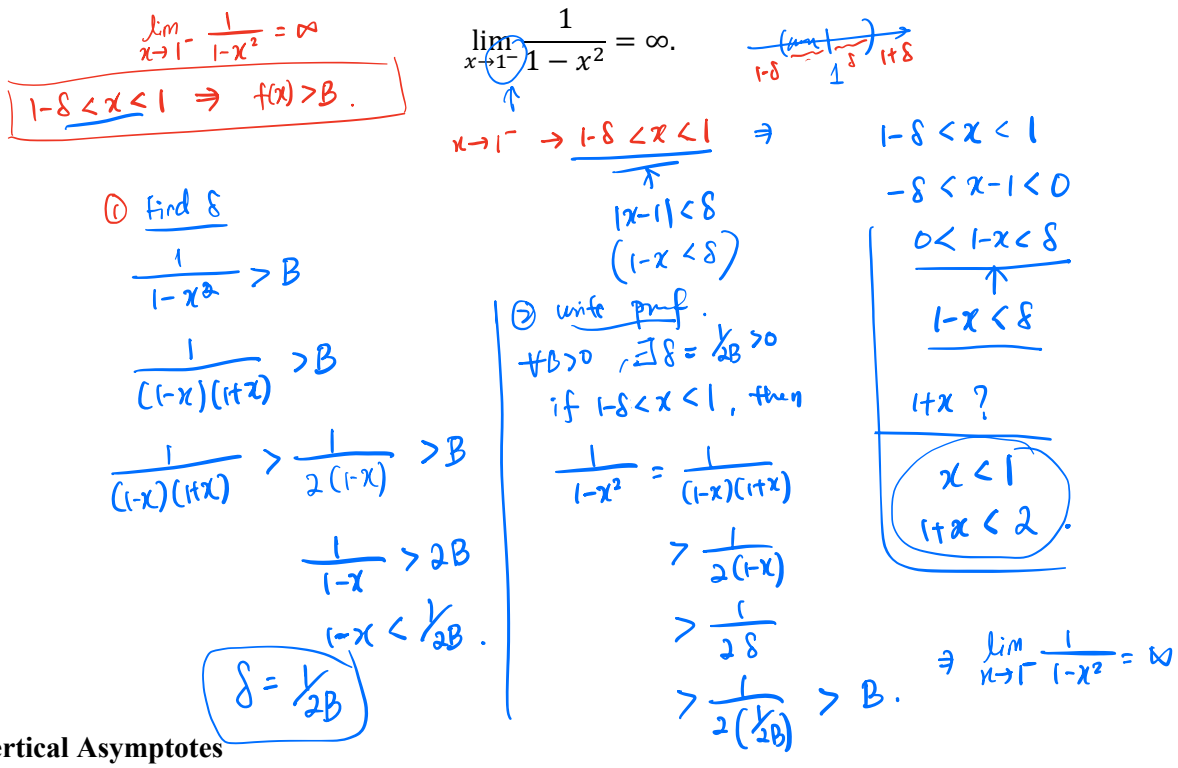
Therefore, by definition

$$\lim_{x\to 0}\frac{1}{x^2}=\infty.$$

Example:

Use formal definition to prove the limit statement

$$\lim_{x\to 1^-}\frac{1}{1-x^2}=\infty.$$



Vertical Asymptotes

Definition. *Vertical Asymptotes*

A line $x=a$ is a vertical asymptote of the graph of a function $y=f(x)$ if either

$$\lim_{x\to a^+}f(x)=\pm\infty \quad or \quad \lim_{x\to a^-}f(x)=\pm\infty$$

Example. *Looking for Asymptotes*

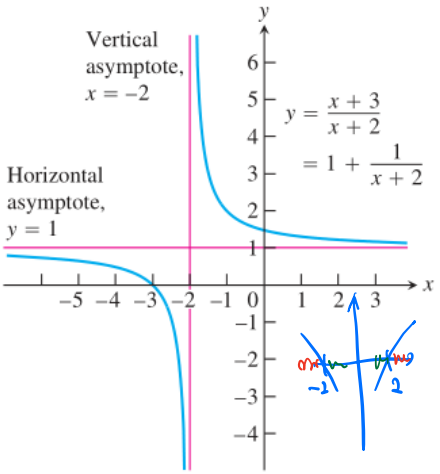
Find the horizontal and vertical asymptotes of the curve

$$y=\frac{x+3}{x+2}.$$

Solution. We are interested in the behavior as $x\to\pm\infty$ and as $x\to -2$, where the denominator is zero. We note that

$$y=\frac{x+3}{x+2}=1+\frac{1}{x+2}.$$

As $x\to\pm\infty, y\to 1$ and as $x\to -2^-$, $y\to -\infty$, as $x\to -2^+, y\to\infty$. Thus, $y=1$ as a horizontal asymptote and $x=-2$ as a vertical asymptote.



Example.

Find the horizontal and vertical asymptotes of the curve

$$f(x) = -\frac{8}{x^2 - 4}.$$

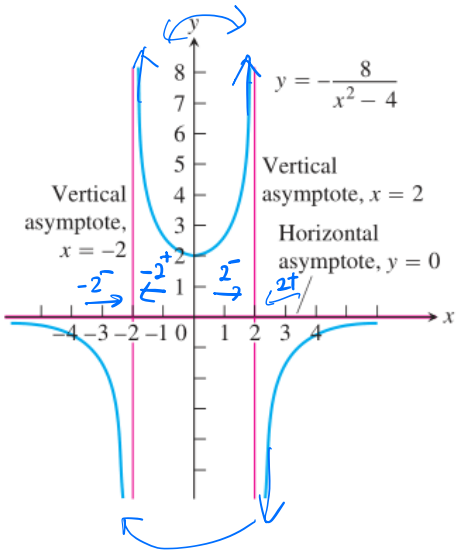
Solution. We are interested in the behavior as $x \to \pm\infty$ and as $x \to \pm 2$, where the denominator is zero. We note that

$$y = -\frac{8}{x^2 - 4} = -\frac{8}{(x-2)(x+2)}.$$

f is an even function of x, so its graph is symmetric with respect to the y-axis.

(a) As $x \to \pm\infty, y \to 0$. Then the line $y = 0$ is a horizontal asymptote of the graph to the right and to the left.

(b) As $x \to \pm 2$. Since $\lim_{x \to 2^+} f(x) = -\infty$ and $\lim_{x \to 2^-} f(x) = \infty$. Then the line $x = 2$ is a vertical asymptote both from left and right. By symmetry, the same holds for the line $x = -2$.



Example: The curve

$$y = \sec x = \frac{1}{\cos x} \quad \textit{and} \quad y = \tan x = \frac{\sin x}{\cos x}$$

both have vertical asymptotes at odd-integer multiples of $\frac{\pi}{2}$.

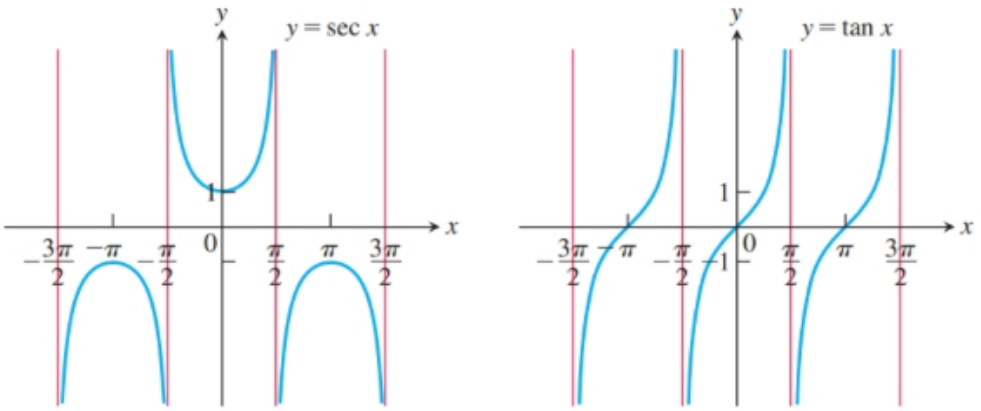


FIGURE 2.52 The graphs of sec x and tan x have infinitely many vertical asymptotes (Example 17).

Dominant Terms

Example:

$$y = \frac{x^2 - 3}{2x - 4} = \left(\frac{x}{2} - 1\right) + \left(\frac{1}{2x - 4}\right)$$

This tells us immediately that

For x large: $f(x) \to \frac{x}{2} + 1$ as $\frac{1}{2x-4} \to 0$

For x near 2: $f(x) \approx \frac{1}{2x-4}$

We say $\frac{x}{2} + 1$ **dominates** when $x \to \pm\infty$, and we say $\frac{1}{2x-4}$ **dominates** when $x \to 2$. Dominant terms like these help us predict a function's behavior.

Example:

Graph the rational function, include the graphs and equations of asymptotes.

1. $y = \frac{x^2-1}{2x+4}$

2. $y = \frac{2x}{x+1}$

Continuity

Any function $f(x)$ whose graph can be sketched over its domain in one continuous motion without lifting the pencil is an example of a continuous function. In this section we investigate more precisely what it means for a function to be continuous.

Definition. *Continuous at a Point.*

Interior point: A function $y = f(x)$ is **continuous at an interior point *c*** of its domain if

$$\lim_{x \to c} f(x) = f(c).$$

Endpoint: A function $y = f(x)$ is **continuous at a left endpoint *a*** or is **continuous at a right endpoint *b*** of its domain if

$$\lim_{x \to a^+} f(x) = f(a) \quad \text{or} \quad \lim_{x \to b^-} f(x) = f(b), \text{ respectively.}$$

- We say that a function is ***continuous over a closed interval*** $[a, b]$, if it is right-continuous at a, left-continuous at b, and continuous at all interior points of the interval.

- This definition applies to the infinite closed intervals $[a, \infty)$ and $(-\infty, b]$ as well, but only one endpoint is involved.

- If a function f is not continuous at a point c, we say that f is **discontinuous** at c and c is a point of **discontinuity** of f. (Note, c need not be in the domain of f).

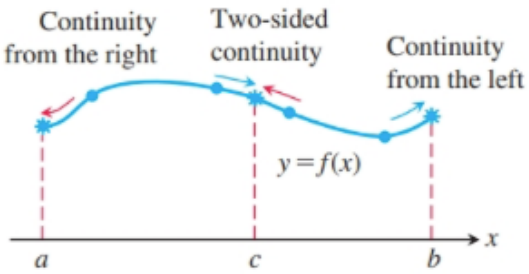
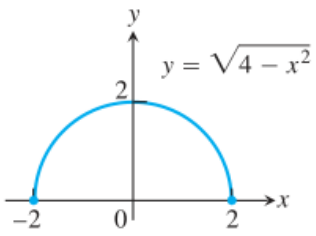


FIGURE 2.56 Continuity at points a, b, and c.

A function f is **right-continuous (continuous from the right)** at a point $x = c$ in its domain if $\lim_{x \to c^+} f(x) = f(c)$. It is **left-continuous (continuous from the left)** at a point $x = c$ in its domain if $\lim_{x \to c^-} f(x) = f(c)$.

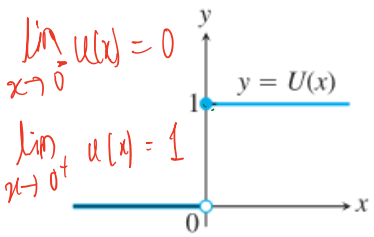
Example. *A Function Continuous Throughout Its Domain*

The function $f(x) = \sqrt{4 - x^2}$ is continuous at every point of its domain, $[-2,2]$, including $x = -2$, where f is right-continuous, and $x = 2$, where f is left-continuous.



Interior point
f continuous (-2,2)
$\lim_{x \to c} f(x) = f(c)$ $c \in (-2,2)$
endpoint [-2,2]
left endpoint -2
$\lim_{x \to -2^+} f(x) = f(-2)$
2 right endpoint
$\lim_{x \to 2^-} f(x) = f(2)$

Example. *The Unit Step Function Has a Jump Discontinuity*



$\lim_{x \to 0} U(x) = 0$
$\lim_{x \to 0^+} U(x) = 1$
$x < 0$ $f(x) = 0$
$x \geq 0$ $f(x) = 1$

The Unit Step function $U(x)$ is right-continuous at $x = 0$, but is neither left-continuous nor continuous there.

We called it as *jump discontinuity* at $x = 0$.

discontinuous at $x = 0$.
↳ jump discontinuity.
$x = 0$.

Example. *The Unit Step Function Has a Jump Discontinuity*

integer floor function

The function $y = \lfloor x \rfloor$. It is discontinuous at every integer n, because the left-hand and right-hand limits are not equal as $x \to n$:

$$\lim_{x \to n^-} \lfloor x \rfloor = n - 1 \text{ and } \lim_{x \to n^+} \lfloor x \rfloor = n.$$

We called it as *jump discontinuity.*

Since $\lfloor n \rfloor = n$, the greatest integer function is right-continuous at every integer n (but not left-continuous).

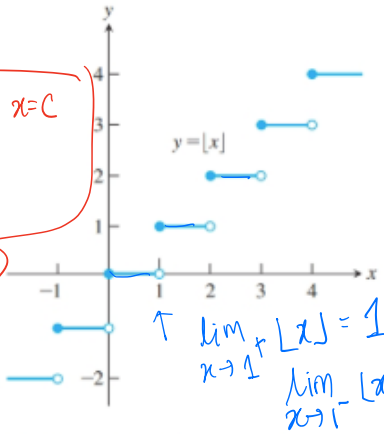


FIGURE 2.59 The greatest integer function is continuous at every noninteger point. It is right-continuous, but not left-continuous, at every integer point (Example 4).

interior
f continuous at $x = c$
$\lim_{x \to c} f(x) = f(c)$
limit exist
f(c)

$\lim_{x \to 1^+} \lfloor x \rfloor = 1$
$\lim_{x \to 1^-} \lfloor x \rfloor = 0$
↳ $\lim_{x \to 1} \lfloor x \rfloor$ doesn't exist
→ discontinuity at $x = 1$
Jump discontinuity.

Continuity Test.

A function $f(x)$ is continuous at $x = c$ if and only if it meets the following three conditions.

1. $f(c)$ exists (c lies in the domain of f)
2. $\lim_{x \to c} f(x)$ exists. (f has a limit as $x \to c$)
3. $\lim_{x \to c} f(x) = f(c)$ (the limit equals the function value).

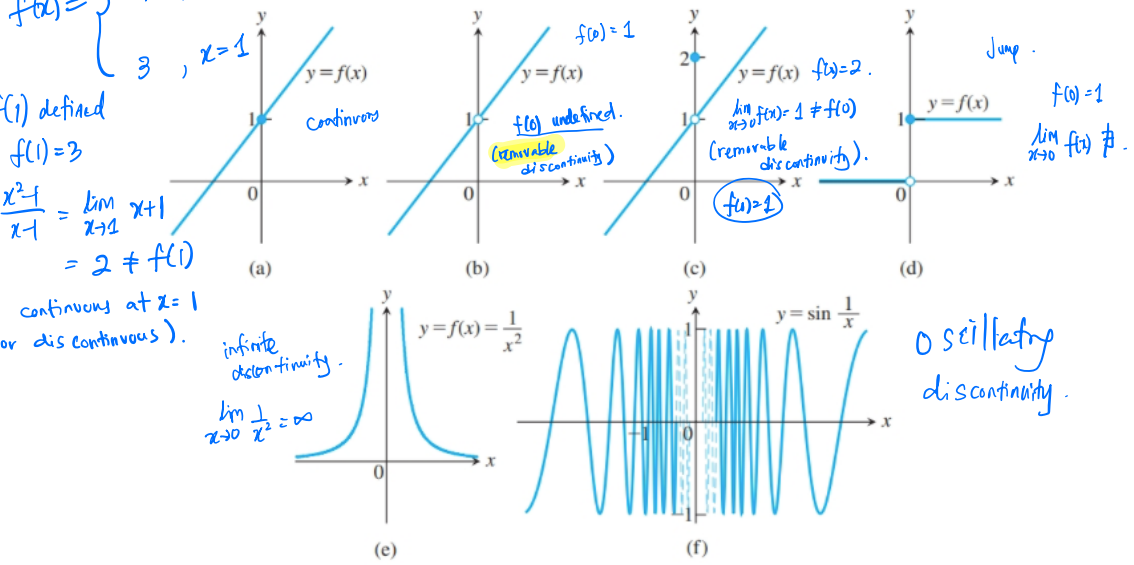


FIGURE 2.60 The function in (a) is continuous at $x = 0$; the functions in (b) through (f) are not.

The figure above is a catalog of discontinuity types.

(a) f is continuous at $x = 0$.

(b) f would be continuous if it had $f(0) = 1$.

(c) f would be continuous if $f(0)$ were 1 instead of 2.

(b) and (c): **removable discontinuity**.

(d) $\lim_{x \to 0} f(x)$ does not exist, and there is no way to improve the situation by changing f at 0.

(d): **Jump discontinuity**

(e) $f(x) = 1/x^2$ has **infinite discontinuity**.

(f) f has **oscillating discontinuity**.

Continuous Functions

A function is *continuous on an interval* if and only if it is continuous at every point of the interval.

Example. The semicircle function $f(x) = \sqrt{4 - x^2}$ is continuous on the interval $[-2,2]$ which is its domain.

Note: A continuous function is one that is continuous at every point of its domain. A continuous function need not be continuous on every interval.

f continuous. at (1,2)

A function always has a specified domain, so if we change the domain, then we change the function, and this may change its continuity property as well. If a function is discontinuous at one or more points of its domain, we say it is a discontinuous function.

Example.
1. $y = 1/x$ is not continuous function on $[-1,1]$, but it is continuous on its domain $(-\infty, 0) \cup (0, \infty)$.
2. The function $y = 1/x$ is continuous function because it is continuous at every point of its domain. It has a point of discontinuity at $x = 0$, however, it is not defined there.
3. The identity function $f(x) = x$ and constant functions are continuous everywhere.

Theorem. *Properties of Continuous Functions*
If the functions f and g are continuous at $x = c$, then the following combinations are continuous at $x = c$.

f, g continuous.

1. $f + g$	(Sums)
2. $f - g$	(Differences)
3. $f \cdot g$	(Product)
4. kf, for any number k	(Constant multiples)
5. f/g, $g(c) \neq 0$	(Quotients)
6. $f^{\frac{r}{s}}$, provided it is defined on an open interval containing c, where r, s are integers.	(Powers)

Example. *Polynomial and Rational Functions Are Continuous*

(a) Every polynomial $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ is continuous because $\lim_{x \to c} P(x) = P(c)$.

(b) If $P(x)$ and $Q(x)$ are polynomials, then the rational function $P(x)/Q(x)$ is continuous whenever it is defined (where $Q(c) \neq 0$).

$\frac{P(x)}{q(x)}$, $q(x) \neq 0$.

Example. *Continuity of the Absolute Value Function*

The function $f(x) = |x|$ is continuous at every value of x.

That is, for $x > 0$, we have $f(x) = x$, a polynomial. If $x < 0$, we have $f(x) = -x$, another polynomial. Finally, at the origin, $\lim_{x \to 0} |x| = 0 = |0|$.

The functions $y = \sin x$ and $y = \cos x$ are continuous at $x = 0$. Both functions are continuous everywhere.

$y = \tan x = \frac{\sin x}{\cos x}$, $\cos x \neq 0$.

It follows from Theorem that all six trigonometric functions are continuous wherever they are defined.

For example, $y = \tan x$ is continuous on $\ldots \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \cup \ldots$.

Continuity of Compositions of Functions

Functions obtained by composing continuous functions are continuous.

If $f(x)$ is continuous at $x = c$ and $g(x)$ is continuous at $x = f(c)$, then $g \circ f$ is also continuous at $x = c$.

In this case, the limit of $g \circ f$ as $x \to c$ is $g(f(c))$.

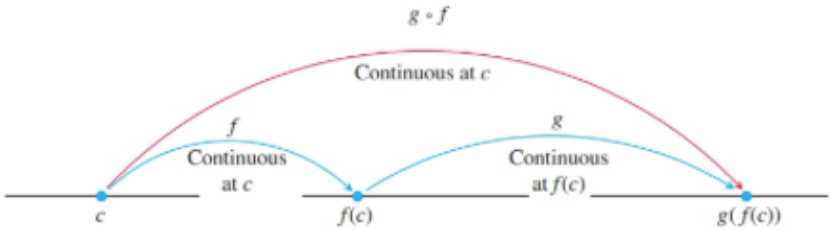


FIGURE 2.62 Compositions of continuous functions are continuous.

Theorem 9 **Compositions of Continuous Functions**
If f is continuous at c, and g is continuous at $f(c)$, then the composition $g \circ f$ is continuous at c.

Intuitively, Theorem 9 is reasonable because if x is close to c, then $f(x)$ is close to $f(c)$, and since g is continuous at$f(c)$, it follows that $g(f(x))$ is close to $g(f(c))$.

Example: Show that the following functions are continuous on their natural domains.

(a) $y = \sqrt{x^2 - 2x - 5}$

(b) $y = \left(\frac{x^{2/3}}{1+x^4}\right)$

(c) $y = \left|\frac{x-2}{x^2-2}\right|$

(d) $y = \left|\frac{x \sin x}{x^2+2}\right|$

Theorem 10 **Limits of Continuous Functions**

If $\lim_{x\to c} f(x) = b$ and g is continuous at point b, then

$$\lim_{x\to c} g\big(f(x)\big) = g(b).$$

Example:

$$\lim_{x\to \frac{\pi}{2}} \cos\left(2x + \sin\left(\frac{3\pi}{2} + x\right)\right) = \cos\left(\lim_{x\to\frac{\pi}{2}} 2x + \lim_{x\to\frac{\pi}{2}} \sin\left(\frac{3\pi}{2} + x\right)\right)$$

$$= \cos\left(\lim_{x\to\frac{\pi}{2}} \left(2x + \sin\left(\tfrac{3\pi}{2} + x\right)\right)\right)$$

$$= \cos\left(\lim_{x\to\frac{\pi}{2}} (2x) + \lim_{x\to\frac{\pi}{2}} \sin\left(\tfrac{3\pi}{2} + x\right)\right)$$

$$= \cos\left(2\left(\tfrac{\pi}{2}\right) + \sin\left(\lim_{x\to\frac{\pi}{2}} \left(\tfrac{3\pi}{2} + x\right)\right)\right)$$

$$= \cos\left(\pi + \sin\left(\tfrac{3\pi}{2} + \tfrac{\pi}{2}\right)\right)$$

$$= \cos\left(\pi + \sin(2\pi)\right)$$

$$= \cos(\pi + 0)$$

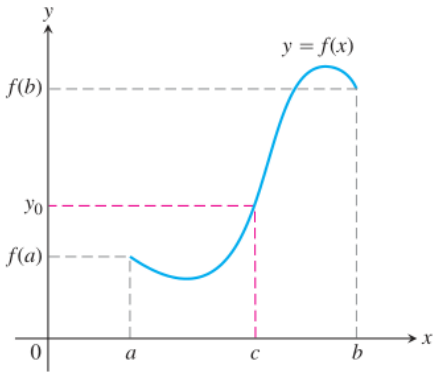
$$= \cos(\pi) = -1.$$

Intermediate Value Theorem for Continuous Functions

A function is said to have the **Intermediate Value Property** if whenever it takes on two values, it also takes on all the values in between.

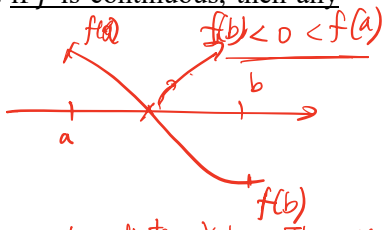
Theorem. *The Intermediate Value Theorem for Continuous Functions*

A function $y = f(x)$ that is continuous on a closed interval $[a, b]$ takes on every value between $f(a)$ and $f(b)$. In other words, if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.



Geometrically, the **Intermediate Value Theorem** says that any horizontal line $y = y_0$ crossing the y-axis between the numbers $f(a)$ and $f(b)$ will cross the curve $y = f(x)$ at least once over the interval $[a, b]$.

A Consequence for Root Finding: We call a solution of the equation $f(x) = 0$ a **root** of the equation or **zero** of the function f. The Intermediate Value Theorem tells us that if f is continuous, then any interval on which f changes sign contains a zero of the function.



By intermediate Value Theorem,
$f(b) < 0 < f(a)$
$\exists\, c \in (a,b)$ s.t.
$f(c) = 0$.
$\Rightarrow$ f has at least a root in (a,b)

Example. Show that there is a root of the equation

$$x^3 - x - 1 = 0$$

between 1 and 2.

$f(x) = x^3 - x - 1$
f is a polynomial, f continuous everywhere.
$f(1) = 1^3 - 1 - 1 = -1 < 0$
$f(2) = 2^3 - 2 - 1 = 5 > 0$
$\Rightarrow \quad -1 < 0 < 5$
The intermediate Value Theorem,
$\exists\, c \in (1,2)$ s.t.
$f(c) = 0 \Rightarrow$ f has at least a root in $(1,2)$.

Example. Use the Intermediate Value Theorem to prove that the equation

$$\sqrt{2x+5} = 4 - x^2$$

has a solution.

$\sqrt{2x+5} = 4 - x^2 \Rightarrow \sqrt{2x+5} - 4 + x^2 = 0$.
$f(x) = \sqrt{2x+5} + (x^2 - 4)$
f continuous $[-5/2, \infty)$.
$f(0) = \sqrt{5} + 0 - 4 < 0$.
$f(2) = \sqrt{9} + 4 - 4 = 3 > 0$.
$f(0) < 0 < f(2)$
By Intermediate Value theorem,
$\exists\, c \in (0,2)$ s.t.
$f(c) = 0$
$\Rightarrow$ f has a root
$\Rightarrow$ equation has a solution.

$f(x) = 4x^3 - 6x^2 + 3x - 2$

$f(1) = 4 - 6 + 3 - 2 = -1 < 0$

$f(2) = 4(2^3) - 6(2^2) + 3(2) - 2 = 12 > 0$

f continuous in $[1,2]$

$\Rightarrow$ By Intermediate Value Theorem, f has at least 1 root in $(1,2)$.

$\Rightarrow$ equation has a solution in $(1,2)$

Example. Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

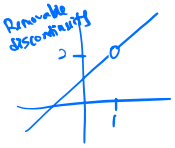
between 1 and 2.

Solution. We let $f(x) = 4x^3 - 6x^2 + 3x - 2$. We note that

$$f(1) = 4 - 6 + 3 - 2 = -1 < 0,$$

$$f(2) = 32 - 24 + 6 - 2 = 12 > 0.$$

Thus, $f(1) < 0 < f(2)$, that is, $y_0 = 0$ is a number between $f(1)$ and $f(2)$. Since f is a polynomial, thus, f is continuous. So, by Intermediate Value Theorem, there is a number c between 1 and 2 such that $f(c) = 0$. This implies that the equation $4x^3 - 6x^2 + 3x - 2 = 0$ has at least one root c in the interval (1,2).



$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1} (x+1) = 2$

$f(1)$ undefined

$f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$

Continuous Extension to a Point

Sometimes the formula that describes a function f does not make sense at a point$x = c$. It might nevertheless be possible to extend the domain of f to include$x = c$, creating a new function that is continuous at $x = c$.

For example, $y = f(x) = \frac{\sin x}{x}$ is continuous at every point except $x = 0$, since $x = 0$ not in domain.

Since $\lim_{x\to 0} \frac{\sin x}{x} = 1$, we can extend the function's domain to include the point $x = 0$ in such a way that the extended function is continuous at $x = 0$.

$$F(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}.$$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

$f(x) = \frac{\sin x}{x}, \quad x \neq 0.$

$f(0) = \lim_{x\to 0} \frac{\sin x}{x} = 1$

$\Rightarrow f$ continuous.

$F(x)$ is continuous at $x = 0$ because

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 = F(0),$$

it meets the requirements for continuity.

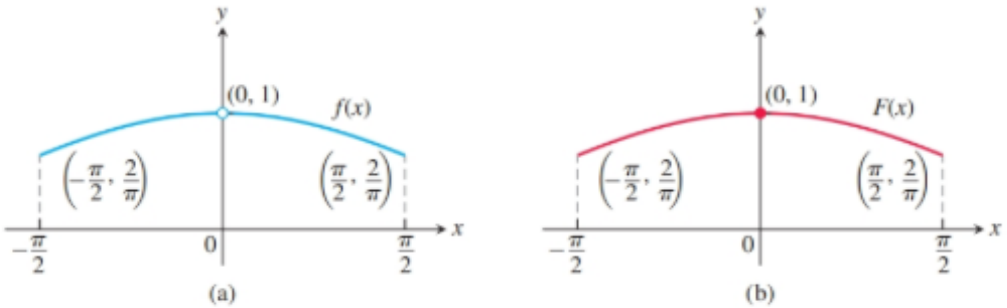


FIGURE 2.67 (a) The graph of $f(x) = (\sin x)/x$ for $-\pi/2 \leq x \leq \pi/2$ does not include the point $(0, 1)$ because the function is not defined at $x = 0$. (b) We can extend the domain to include $x = 0$ by defining the new function $F(x)$ with $F(0) = 1$ and $F(x) = f(x)$ everywhere else. Note that $F(0) = \lim_{x\to 0} f(x)$ and $F(x)$ is a continuous function at $x = 0$.

More generally, a function (such as a rational function) may have a limit at a point where it is not defined.

If $f(c)$ is not defined, but $\lim_{x\to c} f(x) = L$ exists, we can define a new function $F(x)$ by the rule

$$F(x) = \begin{cases} f(x), & x \text{ in domain of } f \\ L, & x = c \end{cases}$$

The function F is continuous at $x = c$. It is called the **continuous extension of f to $x = c$**.

For rational functions f, continuous extensions are often found by *canceling common factors* in the numerator and denominator.

Example: Show that

$$f(x) = \frac{x^2 + x - 6}{x^2 - 4}, x \neq 2$$

has a continuous extension to $x = 2$, and find that extension.

$f(2) = ?$

$$\lim_{x\to 2} f(x) = f(2)$$

$$\lim_{x\to 2} \frac{x^2+x-6}{x^2-4} = \lim_{x\to 2} \frac{(x+3)(x-2)}{(x-2)(x+2)}$$

$$= \lim_{x\to 2} \frac{x+3}{x+2}$$

$$= 5/4$$

$f(2) = 5/4.$

Example: For what value of b is

$$g(x) = \begin{cases} x, & x < -2 \\ bx^2, & x \geq -2 \end{cases}$$

continuous at every x ?

$$\lim_{x\to -2^-} g(x) = \lim_{x\to -2^+} g(x) = g(-2)$$

$$\lim_{x\to -2^-} g(x) = \lim_{x\to -2^-} x = -2.$$

$$\lim_{x\to -2^+} g(x) = \lim_{x\to -2^+} bx^2 = 4b.$$

$$\Rightarrow 4b = -2$$

$$\Rightarrow b = -1/2$$